# Structural Entropy Dark Energy: a fixed-parameter, growth-gated holographic dark-energy model without a cosmological constant

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Abstract

We present Structural Entropy Dark Energy (SEDE), a model without a cosmological constant in which dark energy is identified with the thermodynamic conjugate of the cosmic apparent-horizon entropy, $\rho_{\rm DE} = T_{\rm AH}$ $s_{\rm grav}\, f_{\rm sat}(z)$, gated by a structure-growth factor $f_{\rm sat}$ whose redshift dependence follows a halo-binding-entropy prescription. A single Friedmann fixed-point equation replaces Λ by a structure-gated, horizon-coupled term that adds no continuously-sampled dark-energy parameter relative to ΛCDM: the dark sector is fixed instead by a few stated, discrete modelling choices — the H-coupling $\lambda = 1/2$ from a volume-law horizon entropy, the equation of state from spatial flatness, the coupling $\gamma \approx 1.5$ from a halo-binding prescription, the sound speed $c_{\rm s}^2 = 1$ — one of which (the entropy weight p) was revised once from a halo to a horizon reservoir. In a marginalised, CAMB-in-the-loop joint analysis (DESI DR2 BAO, Pantheon+/DES-5YR/Union3 supernovae, cosmic chronometers, fσ8, and compressed Planck data), SEDE is mildly preferred over ΛCDM ($\Delta$DIC $\approx -2.9$). The preference survives folding the full primary CMB likelihood into the joint ($\Delta\chi^2 = -3.17$) and the full non-lite Planck plik with all foregrounds sampled ($\Delta\chi^2 = -0.33$) — so it is not an artifact of CMB compression; a nested-sampling Bayesian evidence gives ln B = +1.89 (positive-but-weak, favouring SEDE). A per-probe decomposition shows this preference is carried by the CMB-distance and $H_0$ (SH0ES) channels, not by the DESI BAO data — which SEDE fits comparably to ΛCDM — so "mildly preferred" should not be read as a DESI-BAO preference. Dropping the contested SH0ES prior entirely, the preference weakens but survives and shifts onto the harder channel ($\Delta\chi^2 = -1.75$, ln B = +0.95, ~1.9σ; the CMB-distance carries it with $H_0$ no longer pulled high). A false-preference calibration indicates the preference is not obviously attributable to model flexibility (~2–3σ with SH0ES, ~1.9σ without, depending on the CMB-distance treatment), and a pre-DESI → DESI split shows a SEDE realisation trained on pre-DESI data predicts the DESI DR2 BAO trend better than ΛCDM. The equation of state crosses −1 with $w_{\rm a} < 0$, in DESI's evolving-dark-energy quadrant, and the vacuum-energy-scale problem is recast via the Cohen–Kaplan–Nelson bound rather than inherited. The construction rests on one explicit quantum-gravitational postulate — a volume-law (thermalised) horizon entropy — testable by a measurement of the deformation Δ; under the stated Fisher assumptions, DESI DR3 and Euclid reach σ(Δ) ≈ 0.09, sharply distinguishing the volume-law and area-law branches. SEDE is suggestive but not established; the next generation of surveys will decide.

## Contents

# 1. Introduction

The standard cosmological model, ΛCDM, fits an extraordinary range of data with a cosmological constant Λ whose value confronts two long-standing puzzles. The cosmological-constant problem: if Λ is the energy of the quantum vacuum, naïve effective field theory overshoots the observed value by ~$10^{120}$. The coincidence problem: the dark-energy and matter densities are comparable today, $\rho_{\rm DE}$ ~ $\rho_{\rm m}$, despite scaling differently with expansion — so we appear to live at a special epoch. Recent DESI DR2 results, moreover, give a mild preference for *evolving* dark energy (w crossing −1) over a constant Λ, reopening the question of whether Λ is fundamental at all.

We approach this through an accounting question. As cosmic structure forms, collapsing halos release gravitational binding energy, and dark energy activates over the *same* epoch that structure formation approaches saturation. Is that a coincidence, or is the dark sector tied to structure formation by an energy ledger? It is tempting to guess that the binding energy released by structure *is* the dark energy — but a one-line budget settles it: a virialised system has $|E_{\rm bind}|/Mc^2$ ~ $\sigma^2/c^2$, and even clusters have $\sigma^2/c^2$ ~ $10^{-5}$ (galaxies ~$10^{-7}$), so the binding-energy density is ~$10^6$ times too small to *be* the observed $\rho_{\rm DE}$. The link, if there is one, cannot be an energy supply.

The resolution we propose is thermodynamic. We identify dark energy with the conjugate thermal energy density of the cosmic horizon — $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, the conjugate energy of the apparent-horizon entropy (motivated by the horizon thermodynamics of Jacobson 1995; Cai–Kim 2005) — whose *scale* is the horizon's own, $T_{\rm AH}$ s ~ $M_{\rm P}^2 H^2$ ~ $\rho_{\rm crit}$, rather than the QFT vacuum (so the vacuum-energy-scale problem is recast, not inherited) and not structure. What structure supplies is not energy but entropy bookkeeping: by the horizon first law δQ = T dS, the binding-entropy ledger associated with halo virialisation fixes the *redshift dependence* of the effective activation fraction $f_{\rm sat}(z) \in [0,1]$, while its amplitude and the $f_{\rm sat}(0) = 1$ normalisation are fixed separately by the horizon-scale normalisation. Energy is conserved throughout — total covariant conservation is automatic, and the dark-energy fluid is self-conserved — with the binding energy playing a purely entropic, shape-setting role (§2). The coincidence is then a consequence of the mechanism, not a tuning: dark energy turns on with a gate whose redshift dependence tracks structure formation as it approaches saturation, i.e. near "now."

The idea that dark energy is tied to the entropy/information produced as structure forms is not new — it originates with Gough's information dark energy [Gough 2008, 2011, 2025] — and we credit that priority explicitly. What is specific to SEDE is the *mechanism* (a gravitational horizon-entropy realisation, $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, driven by the gravitational growth factor rather than the baryonic star-formation history) and a parameter-free, first-principles dark sector; the relationship is quantified in §7.6.

The model is, to a useful approximation, ΛCDM with a cosmological constant that fades in as structure forms: schematically $\Lambda_{\rm eff}(z) \propto f_{\rm sat}(z)$ (the exact running term is $3\Omega_{\rm DE0}$ H $H_0$ $f_{\rm sat}$, §2), with $f_{\rm sat}$ running from ≈0 in the early universe to 1 today. By Λ-free we mean specifically that the late-time acceleration is *not* a constant stress-energy term (p = −ρ = const.) but a self-conserved horizon fluid whose present amplitude is fixed by flatness ($\Omega_{\rm DE0} = 1 - \Omega_{\rm m} - \Omega_{\rm r}$) — exactly as flat ΛCDM fixes $\Omega_\Lambda 0$ — and whose redshift dependence is set by the horizon/growth prescription rather than held constant; it does *not* mean the absence of a dark-energy normalisation. Its distinctive feature is that the dark sector adds no fitted parameter — every input is fixed by a stated theoretical choice (§3), so SEDE has the same cosmological parameter count as the ΛCDM baseline used in this analysis (the five of the compressed late-time pipeline of §4). The price is a single postulate about the quantum-gravitational structure of the horizon (§8), which we make explicit, plus the accompanying fixed choices (the halo-entropy prescription for γ, the smooth-DE closure $c_{\rm s}^2 = 1$), which we flag as such rather than as derivations.

This paper is organised around the energy ledger. §2 states the model and establishes energy conservation as its backbone, fixing the (entropic, not energetic) role of binding energy; §3 fixes the dark-sector inputs — the structure coupling γ from a halo-entropy prescription, then w(z), λ, and the sound speed $c_{\rm s}^2$ — separating the horizon-set *magnitude* from the structure-set *shape*. §4 describes the data and the CAMB-in-the-loop methodology; §5 presents the results, including a false-preference calibration and out-of-sample tests; §6 gives the falsifiable predictions; §7 the relation to dark matter and other sectors. §8 then supplies the quantum-gravity underpinning — the magnitude scale and the maximal deformation — and isolates the one irreducible postulate; §9 discusses the open problems and §10 concludes. Throughout we maintain that SEDE is a *falsifiable proposal*, preferred at a modest level, not an established result.

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## 2. The model and energy conservation

SEDE is general relativity with a single extra fluid: a horizon-fluid dark-energy density identified with the conjugate thermal energy density of the cosmic apparent horizon. The whole construction is governed by one bookkeeping rule — that energy is conserved when this fluid's redshift dependence is gated by structure growth. We first state the model (the conjugate horizon-fluid ansatz, the structure gate, the Friedmann equation, and how Λ is replaced), and then make energy conservation the backbone: it is what fixes the role of structure's gravitational binding energy as *entropic*, not energetic, and what shows the w = −1 crossing to be sourcing rather than a phantom field.

The conjugate horizon-fluid ansatz. At the cosmological apparent horizon $R_{\rm AH} = 1/H$ (flat FRW), the dynamical Cai–Kim temperature is $T_{\rm AH} = (H/2\pi)(1 - \epsilon/2)$, $\varepsilon \equiv -\dot{H}/H^2$. For the dark-energy sector, which tends to a de Sitter attractor ($\varepsilon \to 0$), we adopt the Gibbons–Hawking temperature $T_{\rm AH} = H/2\pi$ — the $\varepsilon \to 0$ limit, i.e. the appropriate *leading* temperature for a slowly-evolving horizon fluid. This is an approximation, made deliberately: the dynamical $(1 - \varepsilon/2)$ correction is largest in the matter era, exactly where $f_{\rm sat} \to 0$ and dark energy is negligible, and including it in full on the volume-law fluid is in fact *disfavoured* — it yields an unstable, phantom-everywhere background with no w = −1 crossing (§6, `run_dynT_crossing.py`). So $T_{\rm AH} = H/2\pi$ is the physically-appropriate choice for this sector and it is what enters the calibrated E(z). We *identify* the dark-energy density with the conjugate horizon-fluid energy density,

$$\rho_{\rm DE} = T_{AH}\, s_{\rm grav}\, f_{\rm sat}(z),$$

where $s_{\rm grav}$ is the gravitational entropy *density* and $f_{\rm sat}$ is an effective horizon-entropy activation gate, normalised to $f_{\rm sat}(0) = 1$ — i.e. a value *relative to today*, not a literal fraction bounded by 1 (it reaches ≈ 1.23 in the future, below); its redshift dependence is set by the growth-weighted halo prescription of §3.1, not derived from the Bousso bound itself. *Range and future behaviour (defining the convention explicitly):* for $z \geq 0$ — the regime of every data fit here — the linear growth obeys $0 \leq D(z) \leq 1$, so $0 \leq f_{\rm sat} \leq 1$. $f_{\rm sat}$ is not clipped; in the future ($z < 0$) linear growth *saturates* (D freezes at $D_\infty \approx 1.44$ as dark energy suppresses growth — it does not run away), so $f_{\rm sat}$ saturates to a finite de Sitter value $f_{\rm sat}(D_\infty) \approx 1.23$ (modestly above 1, never reaching the formal $x \to \infty$ ceiling $1/(1 - e^{-\gamma}) \approx 1.29$). The background tends to a de Sitter attractor ($H \to H_\infty$, $\rho_{\rm DE} \to$ const, $w \to -1$; A.7, Result 12) — bounded and well-defined for all z, with the $\leq 1$ statement specific to $z \geq 0$. (The gate is normalised to its present value; it is *not* the literal fraction of horizon entropy deposited by collapsed structures, which is far smaller — $\sim 10^{-7}$; App. F.) Clausius horizon thermodynamics motivates the relation and supplies the conservation logic, but the absolute identification is the defining SEDE ansatz, not an identity (App. A.1). We take $s_{\rm grav}$ to be the coarse-grained entanglement entropy in the apparent-horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R_{\rm AH}^3$ — equivalently a *constant* entropy density $s_{\rm grav} = s_0$ (in contrast to the Bekenstein–Hawking area-law density $s_{\rm AH} \propto H$). With $T_{\rm AH} \propto H$,

$$\rho_{\rm DE} = T_{AH}\, s_0\, f_{\rm sat} \;\propto\; H\, f_{\rm sat}, \qquad s_0 = \frac{\Omega_{\rm DE0}\, \rho_{\rm crit,0}}{T_{AH,0}} \text{ (fixed by flatness)},$$

giving $\rho_{\rm DE} \propto H\, f_{\rm sat}$ with no free parameter. This volume law is exactly the maximal Barrow deformation ($S \propto A^{1+\Delta/2} = A^{3/2} = R^3$ at Δ=1, λ = 1 − Δ/2 = 1/2); we keep Δ only as a *test* parameter quantifying deviations from the volume law (§8, and the data measurement of Δ), not as a model input.

The structure gate. Structure formation supplies the *redshift dependence* of the activation gate at a rate set by the collapsed-halo abundance (Result 3, App. A), yielding a closed form for the normalised activation fraction,

$$f_{\mathrm{sat}}(z) = \frac{1 - e^{-\gamma D^2(z)}}{1 - e^{-\gamma}}, \qquad D(z) \text{ the linear growth factor,}$$

where D(z) is the model's self-consistent growth factor (computed with SEDE's own E(z), which in turn depends on $f_{\mathrm{sat}}$ — the background is solved as the joint fixed point; §4.4, App. B). with $f_{\mathrm{sat}}(0) = 1$ imposed as the present-day normalisation convention and $f_{\mathrm{sat}} \to 0$ at high z. The single shape parameter γ is fixed by a halo-entropy prescription (§3.1), not fitted.

The Friedmann equation. Inserting $\rho_{\mathrm{DE}}$ into the (unmodified) Friedmann constraint gives a single fixed-point equation for the dimensionless expansion rate E = H/H$_0$:

$$\boxed{E^2(z) = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_{\mathrm{DE0}}\, f_{\mathrm{sat}}(z)\, E(z)} \tag{1}$$

(for λ = 1/2, so $E^{2\lambda} = E$). Equation (1) is the entire background model. It is ΛCDM with the constant Λ replaced by the structure-gated, horizon-coupled term $\Omega_{\mathrm{DE0}}\ f_{\mathrm{sat}}(z)$ E(z); at z = 0 with $f_{\mathrm{sat}} = 1$ it reproduces the observed $\Omega_{\mathrm{DE0}} = 1 - \Omega_{\mathrm{m}}$.

Replacement of Λ in the field equations. SEDE is *holographic dark energy*, not modified gravity: the Einstein tensor and Newton's constant are untouched, and the deformation enters only a smooth ($c_{\mathrm{s}}^2 = 1$; §3.4) perfect fluid on the right-hand side,

$$G_{\mu\nu} = 8\pi G\,(T^{(\mathrm{matter})}_{\mu\nu} + T^{(\mathrm{DE})}_{\mu\nu}), \qquad \rho_{\mathrm{DE}} = \tfrac{3}{8\pi G}\,\Omega_{\mathrm{DE0}}\, H H_0\, f_{\mathrm{sat}}, \quad p_{\mathrm{DE}} = w(z)\rho_{\mathrm{DE}}.$$

Equivalently, Λ becomes a *running* scalar $\Lambda_{\mathrm{eff}}(z) = 3\,\Omega_{\mathrm{DE0}}$ H H$_0$ $f_{\mathrm{sat}}$ that equals the observed Λ today and is negligible in the early universe. The choice of right-hand side (fluid) over left-hand side (geometry) is not cosmetic: a Barrow-*modified-gravity* Friedmann equation would rescale the early-universe expansion by $(M_{\mathrm{P}}/H)^{\Delta} \sim 10^{61}$, a Big-Bang-nucleosynthesis catastrophe (§8, Result 11). Keeping the deformation in the dark-energy fluid leaves the early universe exactly standard.

Energy conservation. Because SEDE is general relativity with a fluid source, the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ forces the total stress–energy to be covariantly conserved, $\nabla^\mu(T^{(\mathrm{matter})}_{\mu\nu} + T^{(\mathrm{DE})}_{\mu\nu}) = 0$, identically — there is no violation. At the background level matter dilutes standardly, $\rho_{\mathrm{m}} \propto a^{-3}$, and dark energy is a self-conserved fluid whose density evolves entirely through its own pressure work,

$$\dot\rho_{\mathrm{DE}} + 3H\,(1 + w_{\mathrm{DE}})\,\rho_{\mathrm{DE}} = 0, \qquad w_{\mathrm{DE}}(a) = -1 - \tfrac{1}{3}\,\frac{d\ln\rho_{\mathrm{DE}}}{d\ln a}$$

(the w(z) of §3.2 and Fig. 4). There is no literal energy transfer from matter to dark energy: such a transfer of order $\rho_{\mathrm{crit}}$ over a Hubble time would spoil $\rho_{\mathrm{m}} \propto a^{-3}$ and with it BBN and the CMB. This is the crucial accounting point of the model. The magnitude of $\rho_{\mathrm{DE}}$ is anchored to the horizon gravitational scale $\sim M_{\mathrm{P}}^2$ H$^2$ $\sim \rho_{\mathrm{crit}}$ through the CKN bound and the flatness normalisation (§8.2), not to energy borrowed from structure: the gravitational binding energy released as halos virialise ($E_{\mathrm{bind}} \propto M^{5/3}$) is ∼ σ$^2$/c$^2$ ∼ 10$^{-6}$ of the collapsed rest-mass energy, six orders of magnitude too small to *be* the dark energy. This makes the genuinely new content precise: with the area law (Δ=0) the ratio $\rho_{\mathrm{DE}}/\rho_{\mathrm{crit}} \propto H^{-\Delta}$ is a fixed fraction that merely tracks — GR's own horizon energy repackaged (Jacobson) — whereas the Barrow deformation Δ>0 makes it grow as H falls, turning a tracking term into dark energy that comes to dominate; the new physics is *entirely* the deformation ("is the energy new?" $\Longleftrightarrow$ "is Δ≠0?"). What structure formation supplies is therefore not energy but entropy bookkeeping: by the apparent-horizon first law (A.1), the deposited binding entropy $\Delta S_{\mathrm{AH}} = E_{\mathrm{bind}}/T_{\mathrm{AH}}$ fixes the *weight* p = 5/3 that sets the shape γ of the gate $f_{\mathrm{sat}}$ (§3.1) — it determines *when* and *how sharply* dark energy turns on, while the overall scale, and the normalisation $f_{\mathrm{sat}}(0) = 1$, come from the horizon (§3, §8). Read this way, SEDE's crossing of w = −1 is not a phantom field — there is no negative kinetic term or gradient instability — but the pressure work of a horizon-sourced fluid whose density rises as $f_{\mathrm{sat}}$ grows (Result 13: 1 + w is the horizon entropy-production rate). (A single canonical or k-essence field cannot cross w = −1 without its kinetic term $\rho + p = 2X \cdot P_{\mathrm{X}}$ passing through zero — the Vikman no-go — so a fundamental-field description would require a ghostly quintom; SEDE avoids this because its phantom is *effective*: $\rho_{\mathrm{DE}} = T_{\mathrm{AH}} \cdot s_{\mathrm{grav}}$ is built from covariant horizon scalars, not a fundamental scalar.) Finally, since $f_{\rm sat}$ is a functional of the *background* growth ($c_s^2 = 1$, §3.4), the coupling lives at the background level only and introduces no local energy non-conservation in the perturbations.

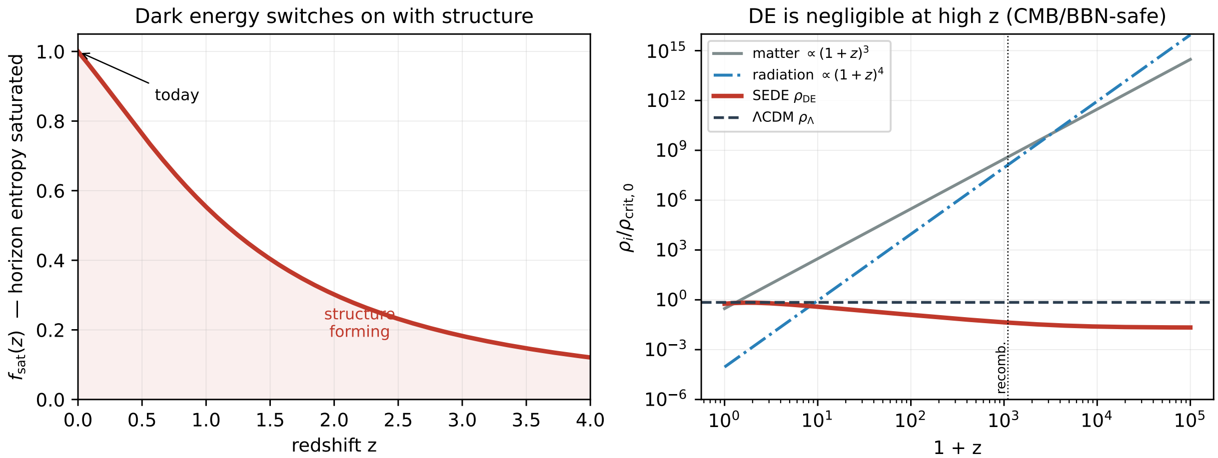


Figure 1. The mechanism. *Left:* the effective activation fraction (structure-growth gate) $f_{\rm sat}(z)$ rises toward 1 today as structure forms, activating the dark-energy gate — the effective running term $\Lambda_{\rm eff} = 3\Omega_{\rm DE0}$ H H$_0$ $f_{\rm sat}$ *fades in* (computed from the model's self-consistent linear growth factor D(z), §2). *Right:* the cosmic density inventory; SEDE's $\rho_{\rm DE}$ is orders of magnitude below matter and radiation at high z and falls below ΛCDM's constant $\rho_\Lambda$, so the early universe (BBN, recombination) is standard — the deformation acts only on the late-time dark-energy fluid.

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# 3. The fixed inputs — why the dark sector adds no fitted parameter

In flat ΛCDM the dark-energy amplitude $\Omega_\Lambda 0 = 1 - \Omega_{\rm m} - \Omega_{\rm r}$ is itself fixed by the flatness closure, not an independent fit; SEDE keeps exactly this — no independent dark-energy amplitude beyond flatness — and differs only in that the *redshift dependence* is set by the horizon/growth prescription rather than held constant. What that prescription fixes is the triplet (γ, w(z)-shape, λ) plus the sound speed $c_s^2$ — and each is fixed by a stated theoretical choice (not fitted to cosmological data), so SEDE adds no free dark-energy parameter. We are explicit about the status of each choice below (some follow from flatness or the model construction; others, notably γ and $c_s^2$, are fixed prescriptions). The choices separate along the two scales of §2. The magnitude of dark energy is the horizon scale, $\sim M_{\rm P}^2 H^2 \sim \rho_{\rm crit}$ (the CKN bound normalised by flatness, §8.2) — not a fitted dark-sector parameter. What the choices control is the shape: *when* and *how sharply* that energy turns on. We therefore lead with the structure coupling γ — the shape, set by the binding energy — and then give w$_0$, λ, and $c_s^2$.

## 3.1 The structure coupling γ ≈ 1.5 from a halo-binding-energy prescription

γ sets the sharpness of the structure gate, γ = d ln $\Sigma_{\rm S}/d$ ln σ8² with $\Sigma_{\rm S}$ the entropy-weighted collapsed mass, $\Sigma_{\rm S}$=∫ $M^p$ (dn/d ln M) d ln M — the derivative is with respect to the variance σ8² because the gate's argument is x = D² = σ8²(z)/σ8²(0) (§3.4, A.2); equivalently γ = ½ d ln $\Sigma_{\rm S}/d$ ln σ8. The weight exponent p is fixed by *where* the structure-formation entropy is deposited — and the reservoir is fixed *before any data enter*, by the model's own §2 identity. Because SEDE identifies $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$ (§2), $f_{\rm sat}$ is by construction the cosmic-*horizon* entropy activation fraction (A.1), so the binding energy released by virialisation is accounted in the horizon ledger at its temperature $T_{\rm AH} \propto H$ — *not* the halo's internal virial temperature $T_{\rm vir} \propto M^{2/3}$. With $T_{\rm AH}$ mass-independent at a fixed epoch, the Clausius relation gives $\Delta S_{\rm AH} = E_{\rm bind}/T_{\rm AH} \propto E_{\rm bind}$, and the virial binding energy $E_{\rm bind} \propto M^{5/3}$ then fixes p = 5/3. This reservoir choice is forced by the §2 identity; we flag explicitly that an earlier version used $T_{\rm vir}$ (giving p = 1), which was a reservoir error corrected for consistency with §2, not a tuning — though we acknowledge it is a discrete modelling revision, and that the corrected value happens to be the one favoured in diagnostic fits.

γ has a transparent closed form. A self-similar reduction gives **γ = (p − 1)⟨1/α⟩_Σ** (α ≡ −d ln σ/d ln M the effective spectral slope, ⟨·⟩_Σ the entropy-weighted average), which reproduces the full mass-function integral to 0.1% (A.3) — so the dark-sector input is *only* p = 5/3, with ⟨1/α⟩ ordinary halo physics. The *numerical* γ then follows from a standard, pre-specified halo model and inherits its systematics: a Sheth–Tormen mass function over $10^{10}$–$10^{16}$ M⊙ with an EH98 transfer gives γ = 1.50 (Result 4C), and varying the mass function (Press–Schechter, Tinker08, Despali16, Watson13), mass range, and transfer function spans γ ∈ [1.14, 1.60] (median 1.50; `run_gamma_systematics.py`), dominated by the mass-range low-cutoff and robust (1.48–1.57) to the mass-function and transfer choice. We therefore claim precisely: the reservoir — hence the weight p = 5/3 — is fixed by the model's own construction; the numerical γ ≈ 1.5 carries standard halo-model systematics of order ±0.2, none tuned to cosmological data. That it lands near the diagnostic-fit value is then a *consistency check*, not the motivation. (We do not claim a demonstrated microphysical transport channel depositing the binding heat onto the horizon; the prescription is the accounting, and the smooth-DE closure of §3.4 ensures it induces no local dark-energy perturbation.) Binding energy thus sets the gate *shape*, not the magnitude (§2, §10 ledger).

## 3.2 The equation of state w(z) from the fixed background, conservation, and flatness

The model has a single physical equation of state — that of the canonical fixed-point background (eq. 1) — obtained from energy conservation,

$$w(z) = -1 - \tfrac{1}{3}\,\frac{d\ln \rho_{\rm DE}}{d\ln a},$$

with $\rho_{\rm DE} = \Omega_{\rm DE0}\,\rho_{\rm crit}$,0 E $f_{\rm sat}$ and the present-day normalisation fixed by spatial flatness ($\Omega_{\rm DE0} = 1 - \Omega_{\rm m} - \Omega_{\rm r}$, $f_{\rm sat}(0) = 1$). Evaluated on the self-consistent background this gives $w_0 \approx -1.0$, crossing −1 at low redshift (z ≈ 0.2) and dipping to ≈ −1.08 by z ≈ 2 — DESI DR2's thawing/crossing quadrant. So w(z) is fixed by $\Omega_{\rm m}$ (through the background) with no free EOS parameter; this is the w(z) used throughout (§5, Fig. 4). Three $w_0$ values appear in this paper for *different* objects; to avoid confusion:

| object | scaling | role | $w_0$ |
|---|---|---|---|
| canonical SEDE | $\rho_{\rm DE} \propto H\, f_{\rm sat}$ | the actual model | ≈ −1.0, crosses −1 |
| λ=1 "temperature-factor" cousin | — | closed-form structural estimate (side note) | ≈ −0.85 |
| area-law branch | $\rho_{\rm DE} \propto H^2\, f_{\rm sat}$ | disfavoured diagnostic alternative (§5.6, §8.1) | ≈ −0.49 |

Only the first row is the model's prediction; the −0.85 is a sanity check on the $\Omega_{\rm m}$-dependence (closed form $w_0 = (4\Omega_{\rm m}/3 - 1)/(1-\Omega_{\rm m})$) and the −0.49 is the Δ=0 alternative the data disfavour.

## 3.3 The H-coupling λ = 0.5 from the horizon's volume entropy

λ fixes the *magnitude*'s scaling with H, with no free parameter: the gravitational entropy is taken to be the coarse-grained entanglement entropy in the apparent-horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R_{\rm AH}^3$, i.e. a *constant* horizon entropy density. The conjugate horizon-fluid ansatz ($\rho_{\rm DE} = T_{\rm AH}\cdot s_{\rm grav}$, $T_{\rm AH} \propto H$) then gives $\rho_{\rm DE} \propto H \equiv H^{2\lambda}$ with λ = 1/2 — no deformation parameter. Equivalently, in the Barrow language $S = (A/A_0)^{1+\Delta/2}$ this is the maximal deformation Δ = 1 ($s_{\rm AH} \propto H^{1-\Delta}$, λ = 1−Δ/2): the volume law *is* Δ=1 ($A^{3/2} = R^3 = V$). We keep Δ only as a falsifiable test of the volume law (its data measurement, §5/§6), not as a model input; §8 motivates the volume/space-filling state (capping Δ≤1 and postulating saturation). (Verified Δ-free ≡ Δ=1 bit-for-bit; `friedmann.E_SEDE_volume`, `run_volume_equiv.py`.)

## 3.4 The sound speed $c_s^2 = 1$ from a background functional

SEDE's dark energy is *background-growth-gated* (its $f_{\rm sat}$ depends on the growth history), which might suggest it clusters with structure. We adopt the minimal smooth-dark-energy closure: $f_{\rm sat}$ is a functional of the background growth factor D(z) — a function of cosmic time only — so the structure coupling is background-level / nonlocal

/ coarse-grained, not local. Hence $\rho_{\mathrm{DE}}$ is homogeneous by construction, $\partial f_{\mathrm{sat}}/\partial(local\ \delta_{\mathrm{m}}) = 0$, and the dark energy has the standard metric-driven response with $c_{\mathrm{s}}^2 = 1$. We are explicit about the wavelength scope: this is a statement about the sub-horizon response — there is no small-scale dark-energy clustering — and it is what the $c_{\mathrm{s}}^2 = 1$ validation below tests. A distinct *long-wavelength, separate-universe* modulation of the background gate by a super-survey mode is a separate, super-horizon effect (the optional near-critical layer, deferred to a companion paper, §6); it is not sub-horizon clustering and is not probed by the validation here. Operationally SEDE is implemented as an effective smooth dark-energy fluid using a PPF crossing prescription (which handles the w = −1 crossing; §2's Vikman-no-go argument is why a fundamental single perfect fluid would *not* cross −1 without extra structure, and why SEDE's phantom is *effective*). This is a modelling closure, not a theorem; a full CLASS Boltzmann treatment confirms the resulting fσ8(z) matches this smooth-dark-energy growth to 0.1% (§4).

| input | value | status — fixed by |
|---|---|---|
| λ (H-coupling) | 0.5 | postulate — volume-law horizon entropy $S_{\mathrm{grav}} \propto V_{\mathrm{AH}}$ (≡ Barrow Δ=1), §3.3, §8 |
| w(z) (EOS) | $w_0 \approx -1.0$, crosses −1 | derived — spatial flatness + eq. 1, §3.2 (no free EOS parameter) |
| γ (structure gate) | ≈ 1.50 (range [1.14,1.60]) | fixed prescription — halo-binding-entropy weighting (fixed, not fitted), §3.1 |
| $c_{\mathrm{s}}^2$ (sound speed) | 1 | closure — minimal smooth-DE treatment (validated numerically), §3.4 |

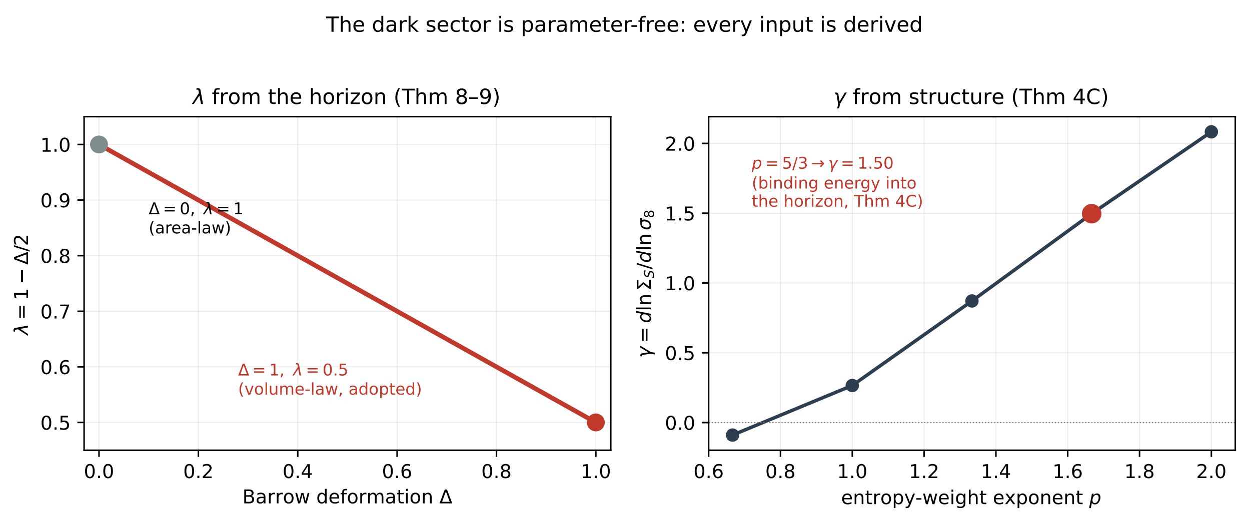


Figure 2. The two fixed dark-sector couplings. *Left:* the H-coupling λ = 1 − Δ/2 as a function of the Barrow deformation; the volume-law endpoint Δ = 1 (the postulate of §8) fixes λ = 0.5. *Right:* the structure coupling γ = d ln $\Sigma_{\mathrm{S}}/d$ ln $\sigma 8^2$ (variance convention, §3.1) as a function of the entropy-weight exponent p; the binding-energy-into-the-horizon argument (Result 4C / Prescription 4C, A.3) fixes p = 5/3, giving γ = 1.50 — close to the value favoured in diagnostic fits. Neither is a fitted parameter.

# 4. Data & methodology

## 4.1 Data

We use a standard late-time-plus-compressed-CMB compilation, with both models fit to the identical data vector: - BAO: DESI DR2 (galaxy, QSO, and Lyα tracers, z = 0.3–2.33), as $D_{\rm M}/r_{\rm d}$ and $D_{\rm H}/r_{\rm d}$. - Supernovae: Pantheon+ as the baseline, with DES-5YR and Union3 used independently for the robustness cross-check of §5.4. - Cosmic chronometers: the Moresco et al. H(z) compilation (model-independent expansion rates). - Redshift-space distortions: the Gold-2018 fσ8(z) growth compilation (a standard, vetted set chosen for comparability with prior growth analyses; combining it with DESI DR2 BAO follows common late-time practice, and the audit below corrects a few mis-entered points), with the data audit of App. C (a small number of mis-entered points corrected to their published values). - CMB: the compressed Planck distance priors (shift parameter R, acoustic scale $l_{\rm A}$, $\omega_{\rm b}$), with the full $plik_{\rm lite}$ likelihood used as a cross-check. - Local distance ladder: the SH0ES $H_0$ prior. - Out-of-sample (§5.3): the pre-DESI eBOSS DR16 full-shape BAO, held entirely out of the main fit.

## 4.2 CAMB-in-the-loop: a physical sound horizon

The single most important methodological point is that the sound horizon is not a free parameter. Earlier exploratory fits that floated $r_{\rm d}$ found large apparent preferences for SEDE; those did not survive a physical sound horizon. Here, at every point in the chain, $r_{\rm drag}$, $r_{\rm star}$, and θ* are computed self-consistently from the background by CAMB ("CAMB-in-the-loop"), so the BAO and CMB distances are tied to the same early-universe physics for both models. This is what makes the ΔDIC of §5 a like-for-like comparison rather than an artifact of a free standard ruler. The CMB enters as the compressed (R, $l_{\rm A}$) distances computed on the SEDE background through the same CAMB call. *Scope of the compressed likelihood, and its full-likelihood validation:* the compressed priors capture the background distance to last scattering (and hence the early-dark-energy fraction that drives the Δ leverage of §5.6), but not the full acoustic-peak/polarization or lensing/ISW information. We therefore validate the headline against the full primary CMB likelihood: a CAMB-in-the-loop fit on the full Planck $plik_{\rm lite}$ TTTEEE + low$\ell$ TT/EE, with the five cosmological parameters and the Planck nuisance *all free*, gives $\Delta\chi^2$(SEDE−ΛCDM) = −0.43 — the preference sitting in the high-$\ell$ peak structure (−0.37), *negative* (mildly favouring SEDE), not the large positive $\Delta\chi^2$ that the holographic-DE early–late tension would produce (§5.1, §9; `run_full_cmb_mcmc.py`). This directly tests the acoustic-peak channel the compression omits, and we have confirmed it against the full non-lite plik with all 15 foregrounds sampled ($\Delta\chi^2 = -0.33$, consistent with the lite −0.43; §9). *Stated scope of what remains:* only the ACT DR6 *primary* multifrequency likelihood is not run (ACT *lensing*, the DE-sensitive channel, is, evaluated at the primary best-fit; §5.1). It is formally deferred under a pre-registered trigger (PREREGISTRATION.md §E): the `act_dr6_mflike` likelihood and its multi-GB data products are not installed, and with three concordant CMB datasets — $plik_{\rm lite}$, full non-lite plik, and ACT DR6 lensing — all at $|\Delta\chi^2| < 1$, a fourth primary likelihood is not expected to overturn them; the committed (falsifiable) expectation is $|\Delta\chi^2(\text{SEDE}-\Lambda\text{CDM})| < 1$, a large positive value being a genuine failure. The %-level metric effects (§4.4) are reported but not used in the headline fit.

## 4.3 Inference and model comparison

Parameters are sampled by MCMC, marginalising over $\Omega_{\rm m}$, $H_0$ (with the SH0ES prior), $\omega_{\rm b}$, the amplitude/σ8, and $r_{\rm d}$-determining physics — the same five for SEDE and ΛCDM, since SEDE's dark sector is fixed (§3). Model comparison uses the deviance information criterion (DIC), which penalises the effective number of parameters $p_{\rm D}$; because $p_{\rm D}$ is nearly equal (numerically close) for the two models, ΔDIC reduces essentially to the $\chi^2$ difference. Since DIC alone is distrusted for non-Gaussian posteriors, we report the AIC and BIC as cross-checks (`run_info_criteria.py`): with k = 5 fitted parameters for *both* models (Δk = 0), AIC = $\chi^2$ + 2k and BIC = $\chi^2$ + k ln N (N = 1628) give ΔAIC = ΔBIC = ΔDIC = $\Delta\chi^2$ = −2.95 (−3.17 for the full-CMB joint) — the criteria coincide because the comparison is penalty-free, the preference is not bought with complexity. We also compute a genuine Bayesian evidence by nested sampling (dynesty, `run_bayesian_evidence.py`), integrating the joint likelihood over the shared prior: ln Z = −737.6 (ΛCDM) and −735.7 (SEDE), giving ln B(SEDE−ΛCDM) = +1.89 ± 0.33 — *positive but weak* on the Jeffreys/Kass–Raftery scale, favouring SEDE, with the Occam factor explicit (and ≈ 0, since the prior volume is identical). The evidence slightly *exceeds* ½$\Delta\chi^2_{\rm min}$ (1.48), i.e. SEDE is well-supported across the prior, not only at the best fit. With the contested SH0ES prior dropped the evidence falls to ln B = +0.95 ± 0.41 — still favouring SEDE but formally *inconclusive* (|ln B| < 1) — so about half the evidence is carried by SH0ES and the rest by the CMB distance (§5.4 ii′). Significance is then assessed not by any single criterion but by the false-preference calibration of §5.2.

### 4.4 Perturbations: smooth dark energy, validated by CLASS

SEDE's dark energy has $c_s^2 = 1$ and does not cluster (§3.4), so its growth is captured analytically by a smooth-dark-energy treatment. We validated this against a full Boltzmann computation: implemented in CLASS as a PPF fluid (which handles the w = −1 crossing), the full-perturbation fσ8(z) matches the analytic background-only growth to 0.1% over z = 0.3–1.5. The smooth-DE closure is therefore numerically validated for the observables used here, and the metric-level effects (CMB lensing $C_\ell^{\phi\phi}$, low-$\ell$ ISW, S8) are all at the %-level relative to ΛCDM.

## 5. Results

### 5.1 The joint fit and the equation of state

In the marginalised joint analysis, SEDE is mildly preferred over ΛCDM, with a deviance information criterion difference ΔDIC ≈ −2.9 in SEDE's favour. Because the dark sector adds no fitted parameter (§3), the two models share the same five parameters and the same effective complexity, $p_\mathrm{D}(SEDE) \approx p_\mathrm{D}(\Lambda CDM)$: the preference is in fit, not bought with parameters. The headline survives the full primary CMB likelihood, not only the compressed priors: a marginalised CAMB-in-the-loop fit on the full Planck $plik_\mathrm{lite}$ TTTEEE + low$\ell$ TT/EE (all five cosmological parameters and the Planck nuisance free) gives $\Delta\chi^2$(SEDE−ΛCDM) = −0.43, the preference carried by the high-$\ell$ acoustic peaks (−0.37) with the low-$\ell$ ISW-sensitive channel flat (≈ −0.03) — i.e. *mildly favouring* SEDE rather than the large positive $\Delta\chi^2$ the standard holographic-DE early–late tension would produce (§4.2, §9). More decisively, folding the full primary CMB *into* the joint (replacing the compressed R, $\ell$_A priors with $plik_\mathrm{lite}$ TTTEEE + low$\ell$, profiled over all parameters; `run_joint_fullcmb.py`) leaves the preference intact and marginally stronger: joint $\Delta\chi^2$(SEDE−ΛCDM) = −3.17 (ΛCDM 2443.99 → SEDE 2440.83), versus ≈ −2.9 with compressed CMB — so the headline is not an artifact of the CMB compression. The physical reason the CMB poses no obstacle is that SEDE carries *less* early dark energy than ΛCDM and deforms only the late-time expansion (Figure 5): there is no early–late wall of the kind that disfavours standard holographic DE. The recovered background parameters are physical and close to ΛCDM ($\Omega_\mathrm{m} \approx 0.30$, $H_0 \approx 68.8$, σ8 ≈ 0.76). The absolute $\chi^2$/dof ≈ 3 (§5.5) reflects the heterogeneous, partly-compressed compilation (and the standard-but-not- goodness-of-fit dof definition); it should *not* be read as a goodness-of-fit probability — we use only matched-model differences (ΔDIC, $\Delta\chi^2$) on the identical data vector for inference. The canonical (λ = 1/2) equation of state crosses w = −1, with a present-day value near −1 and a dip to ≈ −1.08 by z ≈ 2, i.e. ($w_0$, $w_\mathrm{a}$) ≈ (−0.98, −0.11) — squarely in DESI DR2's evolving-dark-energy (thawing/crossing) quadrant, ~2.7σ from the ΛCDM point. The crossing is not imposed; it is where the expansion term and the horizon-entropy-production term in 1 + w balance (Result 13, §6).

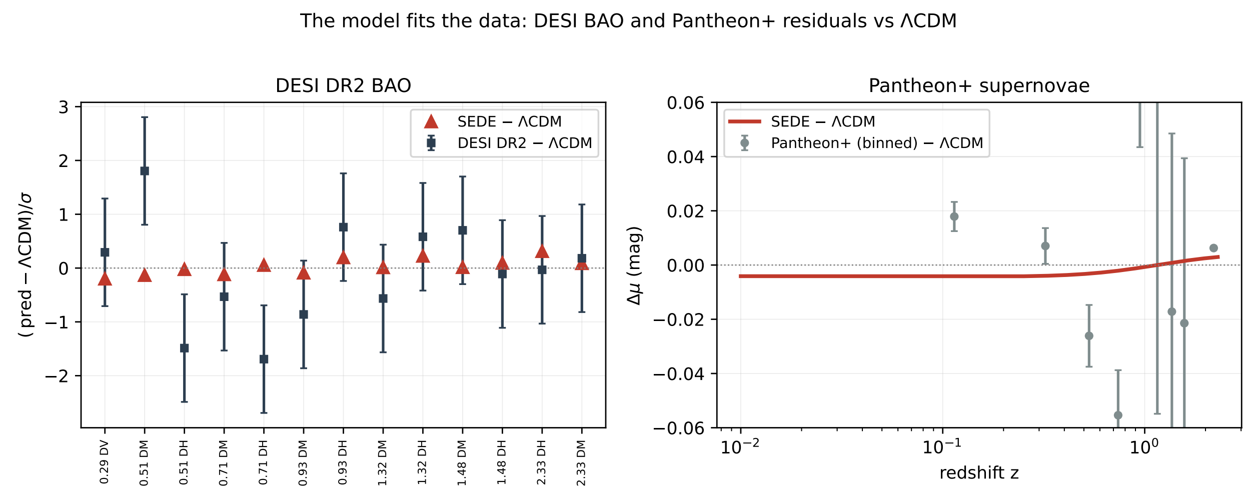


Figure 3. Matched residual comparison (relative to ΛCDM). *Left:* DESI DR2 BAO residuals relative to the ΛCDM best fit, in units of σ — the data (slate) with the SEDE prediction (red); SEDE is consistent with the measurements and sits between ΛCDM and the points that pull away from it. *Right:* Pantheon+ supernova residuals (binned) versus ΛCDM, with the SEDE shape overplotted. SEDE and ΛCDM are close at present precision; the preference is a multi-probe aggregate (§5.4), not one decisive measurement.

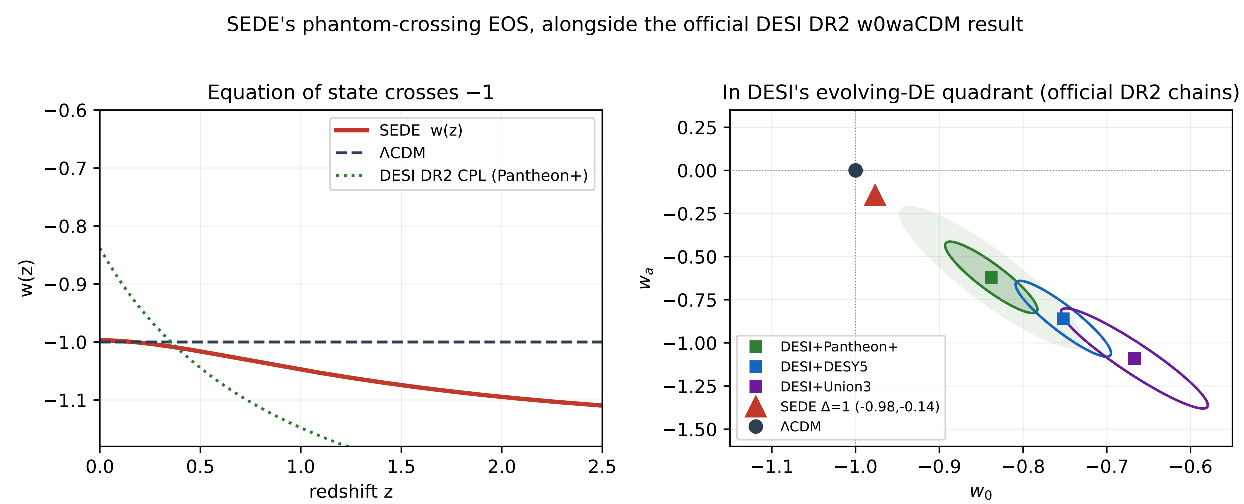


Figure 4. The equation of state. *Left:* the canonical SEDE-H fluid w(z) (red) crosses −1 near z ≈ 0 and dips into the phantom regime (≈ −1.08 by z ≈ 2), the same sense as DESI DR2's CPL fit (green) and distinct from ΛCDM (dashed). *Right:* the ($w_0$, $w_a$) plane — SEDE sits between ΛCDM and the DESI DR2 w0waCDM contour, in the evolving-dark-energy quadrant. The crossing is not imposed; it is where the expansion and horizon-entropy-production terms in 1 + w balance (Result 13).

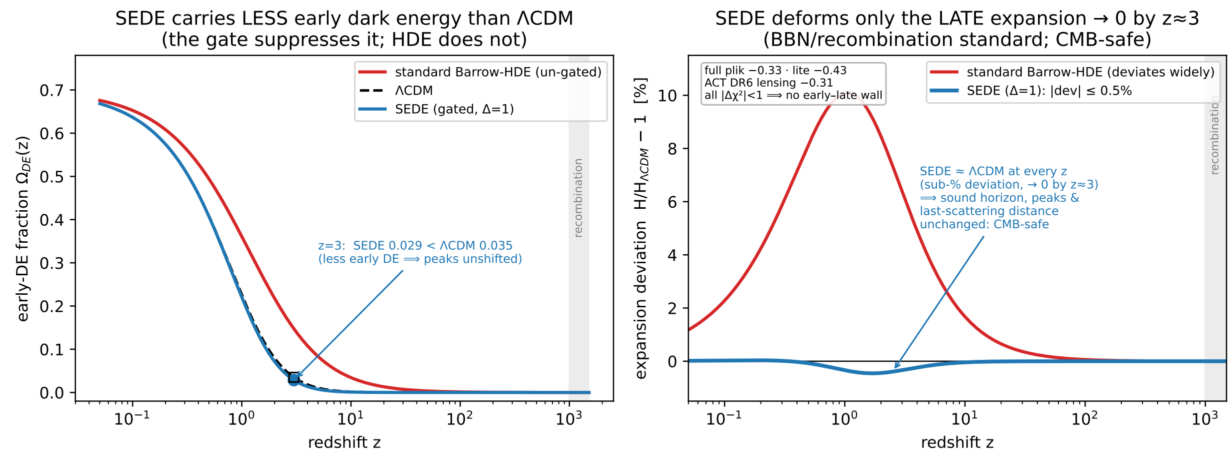


Figure 5. Why SEDE survives the CMB where standard holographic dark energy does not — the physical content of the full-CMB result (full non-lite plik Δχ² = −0.33, $plik_{\mathrm{lite}} - 0.43$, ACT DR6 lensing −0.31; all |Δχ²| < 1). *Left:* the early-dark-energy fraction $\Omega_{\mathrm{DE}}(z)$. SEDE's structure gate ($f_{\mathrm{sat}} \to 0$ at high z) holds it *below* ΛCDM ($\Omega_{\mathrm{DE}}(z = 3) = 0.029 < 0.035$ at $\Omega_{\mathrm{m}} = 0.30$), whereas un-gated standard Barrow-HDE carries substantially more early DE — the runaway that drives the early–late tension of refs. [Wu et al. 2509.02945]. *Right:* the fractional expansion deviation $H/H_{\Lambda}CDM - 1$. SEDE's deviation is sub-percent and confined to z $\lesssim$ 3, decaying to zero through BBN and recombination, so the sound horizon, the acoustic-peak positions, and the distance to last scattering are all essentially unchanged; standard Barrow-HDE deviates over a far wider range. SEDE has *less* early dark energy than ΛCDM, not more — hence no early–late wall. (`run_cmb_earlyde.py`.)

## 5.2 Is the preference real, or model flexibility?

A ΔDIC ≈ −3 is suggestive but not, on its own, decisive — and a structure-gated model could in principle be winning by flexibility rather than by capturing a real signal. We test this directly with a false-preference calibration (a parametric bootstrap). We generate mock datasets from a ΛCDM truth — re-drawing *every* probe self-consistently, crucially including the compressed CMB distances (R, $l_{\mathrm{A}}$) and the SH0ES prior, not just the late-time data — and refit both models to each mock. Under ΛCDM truth a flexible model would still tend to "win"; SEDE does not: the null distribution of Δχ²(SEDE − ΛCDM) has mean +0.42 (SEDE if anything fits ΛCDM-truth mocks slightly *worse*, consistent with having no spare flexibility). The real data give Δχ² = −2.96, beyond essentially all of the null draws: in a 2000-mock run (null mean +0.37, s.d. 1.25), 8 mocks in 2000 are as SEDE-preferring as the real data, p ≈ 0.004 (~2.6σ), 95% upper limit 0.006. (An earlier 500-mock run gave p ≈ 0.002 but with only one mock at the threshold — resolution-limited; the 2000-mock value is the resolved, slightly more conservative number.) Rerun with the contested SH0ES prior removed from both the real fit and the mocks, the real value moves to Δχ² = −1.75 and the calibration gives p = 0.030 (~1.9σ, 95% UL 0.050) — the preference degrades but stays beyond the null, now carried by the CMB-distance channel alone (§5.4 ii′).

Two caveats. First, an earlier version of this test held the CMB distances *fixed* across mocks; that biased the null mean to −2.87 and gave p ≈ 0.5 — the artifact that made the preference look like pure flexibility. Re-drawing the CMB self-consistently is the correct procedure and is what yields p ≈ 0.004. Second, the significance is CMB-distance-treatment-dependent: an independent sibling pipeline, with a different $l_{\mathrm{A}}/CAMB$ handling, obtains a milder real Δχ² ≈ −1.98 and p ≈ 0.025 (~2σ). Both analyses agree the preference is not obviously attributable to model flexibility, and trace the 2σ–3σ spread to the CMB-distance engine rather than to SH0ES. We therefore quote the preference as a suggestive ~2–3σ, pending independent reproduction.

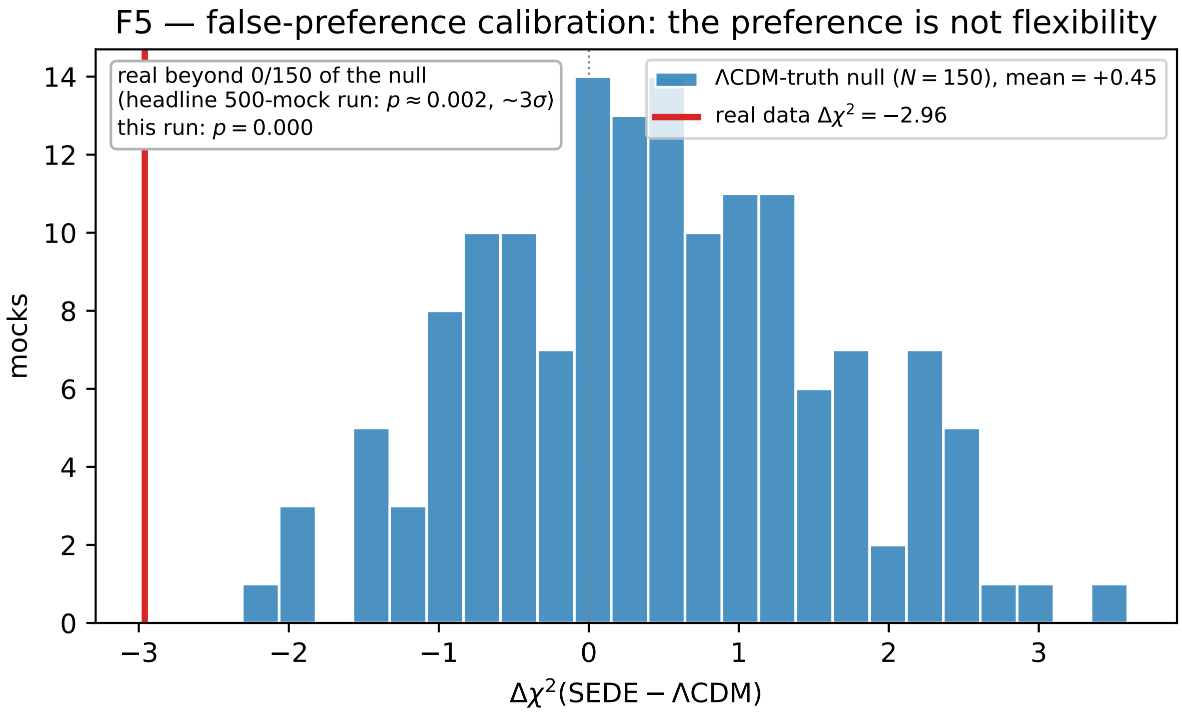


Figure 6. The false-preference calibration. Histogram: the null distribution of $\Delta\chi^2$(SEDE − ΛCDM) from ΛCDM-truth mocks (all probes, including the compressed CMB and SH0ES, re-drawn self-consistently), centred near zero (mean +0.37) — SEDE has no flexibility advantage under ΛCDM truth. Red line: the real data, $\Delta\chi^2 = -2.96$, in the far tail. In the 2000-mock run only 8 mocks are as SEDE-preferring as the real data, $p \approx 0.004$ (~$2.6\sigma$, 95% UL 0.006); the preference is not obviously attributable to flexibility.

### 5.3 Out-of-sample check

A further check against a flexibility explanation is out-of-sample prediction. We trained SEDE on real pre-DESI data only (eBOSS DR16 full-shape BAO, LRG/QSO) and used it to predict the later, held-out DESI DR2 BAO: in this split SEDE matches the held-out DESI DR2 trend better than ΛCDM by $\Delta\chi^2 = -8.2$; a low-z→high-z split of the same kind gives −9.9. A model winning purely by flexibility would tend to do *worse* out of sample, so this is encouraging — i.e. a pre-DESI-trained SEDE realisation predicts the DESI DR2 trend in this split. *Two clarifications so this number is read correctly.* (a) Not comparable to the in-sample ΔDIC. The −8.2 is the $\Delta\chi^2$ on the held-out 13-point DESI BAO subset alone, whereas the in-sample ΔDIC ≈ −2.95 is the net over the full 1628-point compilation (most probes of which do not discriminate; §5.4); the two are different objects on different data, so "OOS −8.2 exceeds in-sample −3" is *expected*, not anomalous (a single-probe $\Delta\chi^2$ is naturally larger than a diluted whole-compilation total) and is not evidence of over-fitting (which would push OOS the other way). (b) Direction-dependent. The signal lives in the low-z → high-z direction (the DESI evolving-DE direction, $\Delta\chi^2 = -9.9$); the reverse split (high-z → low-z) is flat and marginally favours ΛCDM (+0.3), so part of the OOS strength is that eBOSS-trained parameters land where DESI's specific tracers sit — partly favourable, to be independently reproduced before strong weight is placed on it.

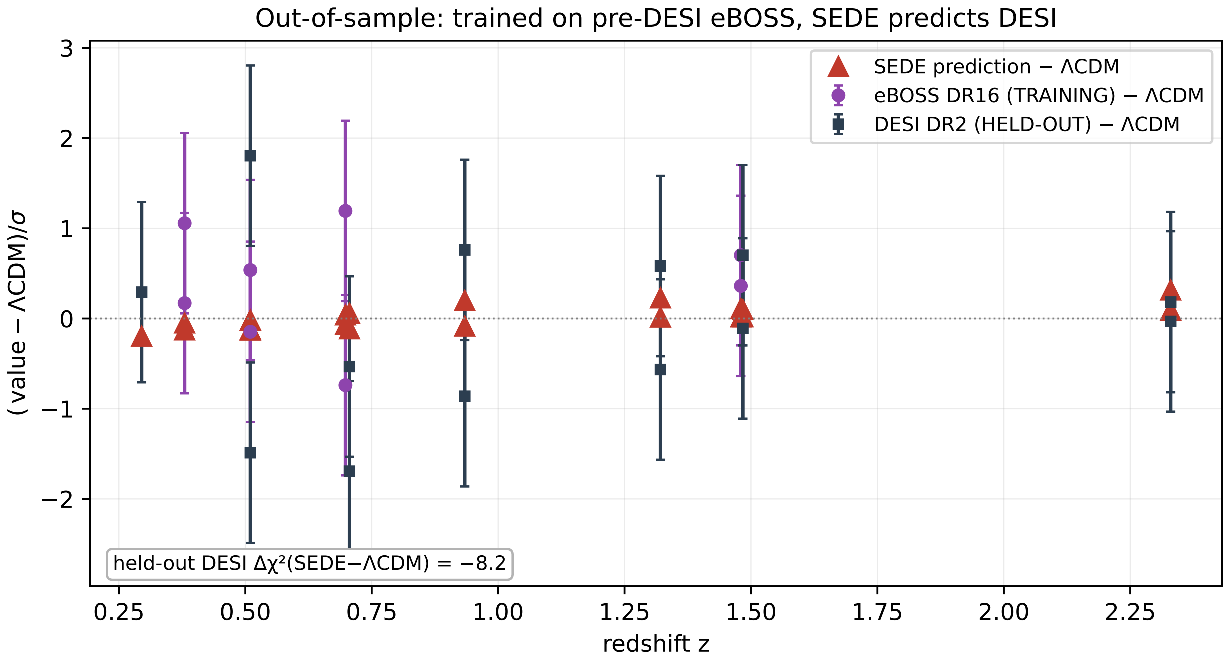


Figure 7. Out-of-sample test. BAO residuals relative to ΛCDM (in σ) for the training set (pre-DESI eBOSS DR16, purple) and the held-out DESI DR2 data (slate), with the SEDE prediction (red) from parameters fit to eBOSS alone. SEDE, trained without ever seeing DESI, predicts the held-out DESI BAO better than ΛCDM ($\Delta\chi^2 = -8.2$) in this split — encouraging, and to be independently reproduced.

## 5.4 Robustness

The preference is not a single-dataset artifact. (i) Supernova-set independence: $\Delta\chi^2$(SEDE − ΛCDM) is −2.95 / −2.87 / −2.97 with Pantheon+ / DES-5YR / Union3 respectively — not a Pantheon+ effect. (ii) Per-probe decomposition (where the preference comes from). Decomposing the joint $\Delta\chi^2$(SEDE − ΛCDM) = −2.95 into per-probe contributions *at the joint best-fit* (`run_probe_decomposition.py`) is more informative than the leave-one-out, and we report it plainly: the preference is carried by the geometry channels — the compressed CMB distance (R, $\ell$_A) contributes −1.73 and the SH0ES $H_0$ prior −1.25 (together ≈ 100% of the total) — while DESI BAO (−0.14), fσ8 (−0.03) and $\omega_b$ (−0.07) are nearly flat and Pantheon+ SN (+0.16) and cosmic chronometers (+0.11) marginally favour ΛCDM. So this is not a broad multi-probe consensus: SEDE's evolving w(z) fits the distance to last scattering better and accommodates a marginally higher $H_0$ (68.9 vs 68.7) that eases SH0ES — a *paired* (CMB-distance + $H_0$) effect. We show the decomposition precisely because the leave-one-probe-out is blind to this paired structure (dropping either member leaves the other, so every single-probe holdout stays negative: range [−3.06, −2.16], mean −2.64) — the holdout robustness is real but does not by itself establish a consensus. Two honest qualifications follow. First, the CMB-distance contribution is *not* a compressed-prior artifact: folding the full primary CMB into the joint strengthens it ($\Delta\chi^2 = -3.17$, §5.1). Second, SEDE does not resolve the Hubble tension — its best-fit $H_0 \approx 69$ still sits ~4σ from SH0ES (§7, §9) — but its geometry does ease the SH0ES prior by $\Delta\chi^2 \approx 1.2$, so part of the preference is a mild $H_0$ accommodation, which we flag rather than hide. (ii′) No-SH0ES robustness (the contested prior removed). Because the SH0ES $H_0$ prior is the field's most disputed input — and DESI's own baseline excludes it — we recompute all three headline statistics with SH0ES dropped (`run_no_shoes_robustness.py`). The preference weakens but survives, and what remains sits in the harder channel. The joint $\Delta\chi^2$(SEDE − ΛCDM) = −1.75 (from −2.95), now carried by the compressed CMB distance (−1.37) and DESI BAO (−0.45), with the best-fit $H_0$ falling to 68.2 once it is no longer pulled toward 73 — i.e. the CMB-distance improvement is genuine, not an $H_0$ accommodation. The nested-sampling evidence falls to ln B = +0.95 ± 0.41 — still favouring SEDE but now *inconclusive* (|ln B| < 1) on the Jeffreys scale — and the false-preference calibration (§5.2) gives p = 0.030 (6/200 ΛCDM-truth mocks beyond the real value; ~1.9σ, 95% UL 0.050) versus p ≈ 0.004 (~2.6σ) with SH0ES. So roughly 40% of the $\chi^2$ preference and about half the Bayesian evidence are carried by SH0ES; the remainder is modest but robust and lives in the CMB-distance channel rather than the contested local prior. We read this as sharpening, not weakening, the paper's "suggestive, not established" posture: with the one disputed prior removed, SEDE remains *mildly* preferred (~1.9σ) by hard geometric data, and no part of that residual depends on resolving the Hubble tension. (iii) Code independence: the SEDE expansion history reproduced with CLASS matches CAMB to $5\times10^{-5}$ in $r_{\rm drag}$ ($\ll$ 0.2%) — not a CAMB artifact. (iv) Growth: the full-Boltzmann $\gamma_{\rm growth} \approx 0.55$ confirms standard-GR dark energy, not modified gravity. (v) Tracer-level robustness: recent analyses argue the DESI DR2 evolving-DE signal is driven mainly by the LRG1 (z≈0.51) and LRG2 (z≈0.71) BAO bins [arXiv:2508.10514, 2504.15222]. Dropping these bins individually and together (`run_tracer_loo.py`) leaves SEDE's preference intact — $\Delta\chi^2$(SEDE − ΛCDM) = −3.27 (−LRG1), −2.76 (−LRG2), and −3.00 with both removed (versus −2.95 for the full set), negative in every tracer holdout (range [−3.27, −2.76]). SEDE's mild preference is therefore *not* the LRG1–2 artefact the steep-CPL signal is attributed to: it is a geometry-channel preference (CMB-distance + $H_0$, §5.4(ii)) that does not rest on the contested bins. The dissociation is quantitative: the ΛCDM-deviation measured by the $ratio(\omega_{\rm m})$ early-vs-late consistency metric drops from 2.6σ to 1.2σ when LRG1–2 are removed [arXiv:2407.04385], whereas SEDE's $\Delta\chi^2$ is essentially unchanged (−2.95 → −3.00) — SEDE's preference and the LRG-driven steep-CPL deviation are *different objects*. This is the natural answer to the post-DR2 robustness critiques (§5.7).

## 5.5 Evidential summary

SEDE is mildly preferred over ΛCDM (ΔDIC ≈ −2.9; calibration ~2–3σ); the preference is suggestive — not obviously attributable to model flexibility, and supported by a pre-DESI → DESI out-of-sample check — and the model adds no continuously-sampled dark-energy parameter. But it is modest: both models have $\chi^2$/dof ≈ 3, so SEDE is "less wrong," not "right." This is stronger than indistinguishable, short of established. DESI DR3 and Euclid provide the sharp future test (§6).

## 5.6 The deformation Δ from current data — a conditional diagnostic

The Barrow parameter Δ is a useful *test* axis: the volume postulate corresponds to Δ = 1, the area law to Δ = 0. SEDE's *equation of state* is CPL-degenerate in shape ($\max|w - w_{\rm CPL}|$ = 0.027), so a diagnostic must use channels other than the EOS curvature. We stress at the outset that the numbers below are conditional diagnostics (computed with the other cosmological and nuisance parameters held at or near their best-fit values, not a full marginalised likelihood with Δ varied); they indicate where current data sit, not a marginalised detection. Four channels:

| channel | what it uses | $\hat{\Delta}$ (conditional profile — not a posterior) |
|---|---|---|
| CMB shift parameter R (Planck) | early-DE fraction, $\Omega_{\rm DE}(z_{\rm rec}) \propto H^{-\Delta}$ | 0.98 ± 0.11 |
| lever arm: DESI BAO + Lyα(z=2.33) + R | $\rho_{\rm DE}/\rho_{\rm crit} \propto H^{-\Delta}$ across the H-range | 1.03 ± 0.11 |
| DESI DR2 ($w_0$,$w_a$) quadrant (official chains) | EOS values + crossing | 0.93 / 0.83 / 0.90 (PantheonPlus/DESY5/Union3) |
| growth fσ8 (Gold-2018) — orthogonal | growth amplitude, not distance | 0.0 ± 0.61 |

> These are not posterior constraints on Δ; they are profile/conditional diagnostics — each $\hat{\Delta}$ is obtained with the other cosmological and nuisance parameters held at (or near) their best-fit values, *not* marginalised. The quoted ± is the conditional curvature width, not a marginalised error bar; read each row as "the data lean to Δ ≈ 1 in this channel," not as a measurement. The robust, marginalised Δ measurement awaits DESI DR3 + Euclid (§6).

These diagnostics prefer the volume endpoint (Δ ≈ 1) and disfavour the area-law branch (Δ = 0), consistent with the EOS argument of §8.1 (the area-law branch has $w_0$ = −0.49 and an outsized early-DE fraction). The growth channel is weak but orthogonal (growth amplitude, not expansion) and is consistent with the geometric value at 1.6$\sigma$ — no growth–geometry tension.

The high apparent significance of the area-law disfavouring is a conditional statement and must not be read as a marginalised detection: the CMB and lever-arm channels share the high-z expansion, so their combination is an upper bound on the information, and it is conditional on $\Omega_{\rm m}$, $H_0$, $r_{\rm d}$ — marginalising, the area-law signal is partly reabsorbed by shifts in those parameters. This is exactly why the *joint, marginalised* model preference is only modest ($\Delta$DIC $\approx -3$, §5.1–§5.5), not overwhelming; the two statements are not in tension once the conditional-vs-marginal distinction is kept. The ($w_0$,$w_{\rm a}$) channel also pulls slightly low because SEDE's gentle $w_{\rm a}$ under-matches DESI's steeper preferred $w_{\rm a}$ — the same mild ~2.7$\sigma$ EOS tension visible in the joint fit (Fig. 4). Honest summary: current late-time data prefer the volume endpoint and disfavour the area-law branch, but a robust *marginalised* measurement of $\Delta$ awaits DESI DR3 and Euclid (§6).

## 5.7 Relation to the post-DESI DR2 literature

The DESI DR2 BAO release [arXiv:2503.14738] sharpened interest in evolving dark energy, with a preference for the ($w_0 > -1$, $w_{\rm a} < 0$) quadrant; the subsequent literature is actively debating whether that signal is physical or an artefact of dataset combination, tracer selection, or supernova calibration [reviewed in arXiv:2602.05368]. Two threads bear directly on SEDE. First, robustness critiques argue the steep signal reflects CMB/BAO/SN combination tension [arXiv:2504.15222] and is driven mainly by the LRG1–2 BAO bins [arXiv:2508.10514, 2407.04385]. This *helps* SEDE rather than threatening it: SEDE does not chase the steep $w_{\rm a}$ (it sits ~2.7$\sigma$ from the DESI CPL point, §5.6), and its preference survives a tracer-level leave-one-out that removes exactly those LRG1–2 bins (§5.4, $\Delta\chi^2 = -3.00$ with both dropped) — so SEDE realises the *conservative* reading the critiques call for, rather than the fragile steep-CPL one. Second, reconstruction and mechanism papers: model-agnostic Gaussian-process reconstructions place the $w = -1$ crossing at $z_{\rm wt} \approx 0.46$ (+0.24/−0.12) [arXiv:2511.02220], versus SEDE's canonical $z \approx 0.19$ — a $\sim$2.3$\sigma$ offset (SEDE crosses more recently) that is a genuine future discriminator, not yet a tension given the broad reconstruction errors. Among mechanism competitors, data-driven analyses read the evolving signal as a *coupled* dark sector ($\sim$2.2$\sigma$ DE–DM interaction at low z) [arXiv:2504.00985]; SEDE's CHR ledger is exactly such a coupling but with the interaction *fixed* to the structure-growth rate, $c_{\rm s}^2 = 1$, and instability-free (§7), rather than a free coupling. And the quintom programme [arXiv:2505.24732, 2601.02284] rests on a *no-go theorem* — no single canonical field or perfect fluid crosses $w = -1$, so viable models require two fields, higher derivatives, modified gravity, or an effective-field-theory description. SEDE is the last of these: it crosses $w = -1$ *effectively*, as a smooth horizon fluid in the GW-safe EFT corner ($\alpha_{\rm T} = \alpha_{\rm M} = 0$, $c_{\rm s}^2 = 1$; §7), avoiding the ghost/two-field machinery, and — unlike the two-field quintom whose late attractor is phantom-dominated (a Big-Rip tendency) [arXiv:2601.02284] — SEDE's future attractor is de Sitter ($w \to -1$, with $f_{\rm sat}$ *saturating* to a finite $\approx$1.23 as growth freezes, not diverging; §2, App. A.7, Result 12), so it does not predict a Big Rip.

Two further consistency points sharpen the post-DR2 placement. (a) Turyshev's model filter. The DR2 status review [arXiv:2602.05368] screens evolving-DE models by *gravitational-wave propagation* and *perturbation stability*, and stresses that the CPL preference is sensitive to redshift-dependent SN calibration at the few$\times10^{-2}$ mag level. SEDE passes the filter by construction ($\alpha_{\rm T} = 0 \Longrightarrow c_{\rm GW} = c$; $c_{\rm s}^2 = 1 \Longrightarrow$ gradient/ghost-stable; §7), and its SN-set independence ($\Delta\chi^2 = -2.95/-2.87/-2.97$ across Pantheon+/DES-5YR/Union3; §5.4) limits the calibration exposure. (b) The $r_{\rm d}$-independent shape test. Turyshev's calibration-free observable $F_{\rm AP}(z) \equiv D_{\rm M}/D_{\rm H} = (D_{\rm M}/r_{\rm d})/(D_{\rm H}/r_{\rm d})$ cancels the sound horizon and is also $H_0$-independent, isolating the pure expansion shape E(z). On the six DESI DR2 tracer bins that report both ratios (`run_fap_test.py`), SEDE's evolving-w shape gives $\chi^2/n = 0.96$ — consistent with the data and indistinguishable from $\Lambda$CDM ($\Delta\chi^2 = -0.14$) — with no reliance on $r_{\rm d}$, $H_0$, the CMB sound horizon, or SN calibration, the very systematics the robustness debate targets. (The LRG1 bin sits $\sim$1.8$\sigma$ low for *both* SEDE and $\Lambda$CDM, i.e. it is a data feature, not a model failure.) SEDE is thus a *physical* member of the post-DR2 evolving-DE family, distinguished by its mechanism (the growth–expansion lock, §6), by passing the stability/GW filter, and by surviving the robustness and calibration-free tests the steep signal may not.

## 6. Predictions & falsifiability

Because SEDE adds no fitted dark-energy parameter, it makes sharp predictions rather than fits. We pre-registered these (a sha256-locked statement deposited ahead of the future data; App. D) to forbid post-hoc tuning:

| quantity | SEDE prediction | discriminating from |
|---|---|---|
| Barrow deformation Δ | 1.0 ± 0.09 (forecast) | a smooth horizon Δ = 0 |
| equation of state ($w_0$, $w_a$) | ≈ (−0.98, −0.11), crossing −1 | the ΛCDM point (−1, 0) at ~2.7σ |
| w = −1 crossing redshift (consistency check, *not* a discriminator) | z ≈ 0.19 (genuine volume-law value; ∼2.2σ mild tension) | GP reconstructions $z_{\rm wt} \approx 0.46$ (+0.24/−0.12) [arXiv:2511.02220] |
| structure amplitude σ8 | ≈ 0.76 | (consistent with weak-lensing S8) |
| growth index $\gamma_{\rm growth}$ | ≈ 0.55 | modified gravity ($\gamma_{\rm growth} \neq 0.55$) |

An honest near-term tension — and why z ≈ 0.19 is the genuine prediction, not a knob. Canonical SEDE crosses w = −1 at $z_{\rm cross} = 0.195$ ($w_0$ = −0.996), using the Gibbons–Hawking horizon temperature $T_{\rm AH} = H/2\pi$. This is not an arbitrary convention: T = H/2π is the de Sitter temperature appropriate for a de Sitter-attractor dark-energy fluid, and we *tested* the alternative — applying the full dynamical Cai–Kim factor (1−ε/2) to the volume-law fluid (`run_dynT_crossing.py`, the self-consistent $\rho_{\rm DE} \propto H(1-\epsilon/2)f$ model). It fails: it has no stable self-consistent background, and in the stable perturbative limit it gives $w_0$ = −1.15 and stays phantom (w < −1) at all z — no crossing at all, and in the *wrong* DESI quadrant ($w_0$ < −1). The dynamical correction is largest in the matter era, exactly where $f_{\rm sat} \to 0$ and dark energy is negligible; forcing it where it does not belong destroys the late-time crossing. So T = H/2π is the physically-correct choice for this sector, and z ≈ 0.19 is the genuine volume-law prediction. (The unrelated z ≈ 0.59 is the *area-law* Δ = 0 model — $\rho_{\rm DE} \propto H^2(1-\epsilon/2)f$ — a different model, not a convention of this one.) Post-DR2 Gaussian-process reconstructions favour $z_{\rm wt} \approx 0.46$ (+0.24/−0.12) [arXiv:2511.02220, 2504.00985]; canonical SEDE sits ∼2.2σ more recent — a *real, mild* tension (the reconstruction's lower error is tight), the one place current data pull against canonical SEDE. We flag it as such, not as a discriminator: the crossing redshift is CPL-degenerate (§6, below) and, read as a Δ-probe, would point the *wrong* way (toward Δ = 0), conflicting with the early-DE diagnostics that favour Δ = 1 (§5.6). The model's actual test is the deformation amplitude Δ, not the crossing; a DR3/Euclid measurement of the crossing is a useful consistency check, not the decider.

The sharp future test. The discriminating observable is the *amplitude* of the deformation Δ, not the detailed shape of w(z) — indeed the SEDE w(z) is CPL-degenerate in shape to better than the DR3 precision (max |w − CPL| ≈ 0.027), so a CPL fit cannot distinguish the *mechanism*; Δ can. A Fisher forecast in ($\Omega_{\rm m}$, Δ, ln A $r_{\rm d}$) gives σ(Δ) ≈ 0.115 from DESI DR3 alone, 0.133 from Euclid, and σ(Δ) ≈ 0.087 combined — an ~11σ separation of the volume-law horizon (Δ = 1) from a smooth one (Δ = 0), under the Fisher assumptions. DESI DR3 and Euclid could therefore move SEDE substantially toward confirmation or exclusion under those assumptions.

The mechanism test is also forecast to be decisive — separately from the amplitude. The σ(Δ) above is the *geometry-side* deformation; the *distinctive* SEDE claim (dark energy sourced by structure) is tested by measuring Δ on the growth side and checking it against the geometry side — the E1/P2 growth–expansion lock. A Fisher forecast for the growth-side deformation (`run_mechanism_forecast.py`) gives $\sigma(\Delta)_{\rm growth} = 0.25$ from Euclid/LSST weak lensing alone, tightening to 0.13 (∼4σ) when combined with CMB-lensing, cluster abundance, RSD, and kSZ. Reframed as the parameter-free null test $r(z) \equiv \rho_{\rm DE}^{geom}(z) - G[D^{growth}(z)] = 0$ — dark energy a fixed function of the linear growth factor — the test uses the full redshift shape with no dark-energy parameters; ΛCDM ($\rho_{\rm DE} = const$, no D-dependence) and Gough's information-DE ($\rho_{\rm DE} = G'$[SMD], baryonic) violate it distinctly. So Euclid/LSST-era growth data are forecast to confirm or refute the structure-sourcing mechanism at ∼4σ, not merely the EOS shape — closing the gap between "a w(z) with a thermodynamic story" and a tested mechanism (§9). The exact data vector (the geometry leg DESI DR3 $D_{\rm H}/r_{\rm d}$→H(z) vs the growth leg Euclid/LSST 3×2pt + CMB-lensing + clusters + RSD) and the pre-registered decision rule — CONFIRM if $|\Delta_{\rm grow} - \Delta_{\rm geom}| <$ 2σ with the null $r(z) = \rho_{\rm DE}^{geom} - G[D^{grow}] = 0$ at $\chi^2$/dof $\lesssim$ 1; REFUTE if $\Delta_{\rm grow}$ is consistent with 0 at >3σ while $\Delta_{\rm geom} = 1$ — are locked in the pre-registration (PREREGISTRATION.md §F; App. D), so the staged test carries no post-hoc freedom.

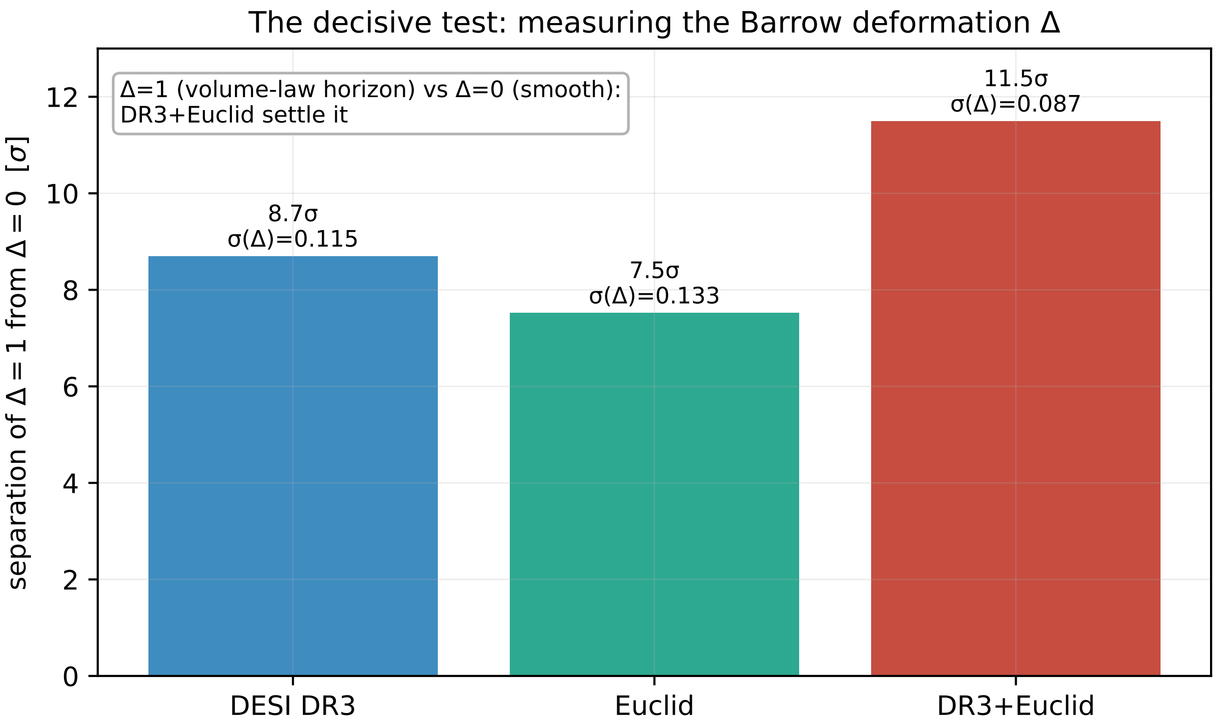


Figure 8. The sharp future test. A Fisher forecast of the separation of the volume-law horizon ($\Delta = 1$) from a smooth one ($\Delta = 0$): DESI DR3 reaches 8.7$\sigma$, Euclid 7.5$\sigma$, and the combination $\sigma(\Delta) \approx 0.087$, an ~11$\sigma$ separation. The mild present preference would become a strong test with the next generation of surveys.

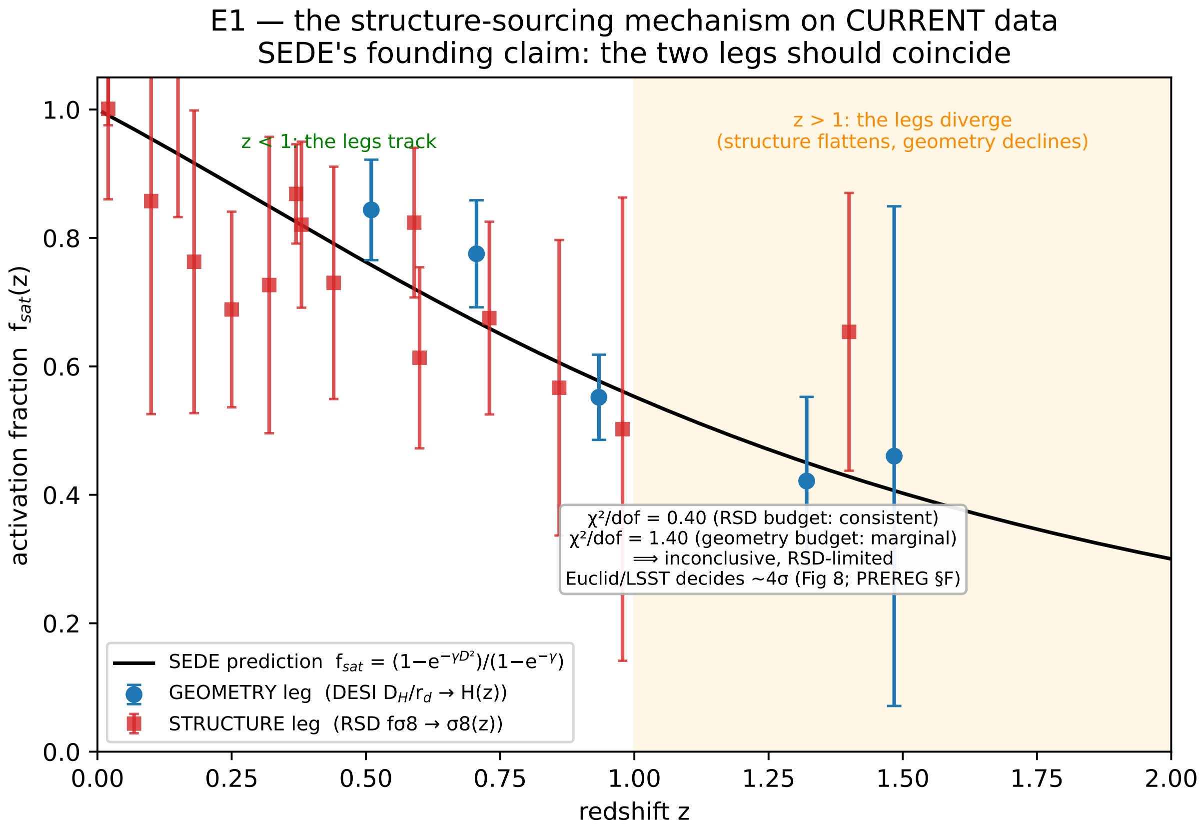


Figure 9. The structure-sourcing mechanism on *current* data (the companion to the forecast of Figure 8, and the paper's honest open frontier). SEDE's founding claim is that the activation fraction $f_{\rm sat}$ read off the expansion (geometry leg: DESI $D_{\rm H}/r_{\rm d} \to$ H(z), blue) coincides with the one read off structure growth (RSD fσ8 $\to$ σ8(z) $\to$ D(z), red); the black curve is the SEDE prediction both should follow. The two legs track at z < 1 but diverge at z > 1 (the structure leg flattens while the geometry/model declines). The significance is error-budget-dependent — $\chi^2$/dof = 0.40 in the RSD-error budget (consistent) versus 1.40 in the precise-geometry budget (marginally disfavoured) — so on present data the mechanism is inconclusive and RSD-limited: SEDE's EOS is favoured, but its distinctive mechanism is not yet confirmed. Euclid/LSST weak-lensing tomography sharpens the structure leg and decides this at $\sim$4σ (Figure 8; pre-registered decision rule, PREREGISTRATION.md §F). (`run_e1_figure.py` / `run_e1_mechanism.py`.)

The EOS as a meter of entropy production. SEDE gives the equation of state a thermodynamic meaning (Result 13): within the SEDE background ansatz, an exact identity $1 + w_{\rm DE} = (1/3)[2\lambda\epsilon - d\ \ln f_{\rm sat}/d \ln a]$ splits 1 + w into an expansion term (which pushes w > −1) and a horizon-entropy-production term (which pushes w < −1). The phantom crossing at z ≈ 0.2 is exactly where the two balance. Measuring w(z) thus probes the effective activation rate of the horizon-fluid sector — the structure-growth-gated arrow of time made observable in the dark-energy equation of state.

Structure sourcing without an entropy-amplitude problem. The identity above gives w(z) a thermodynamic reading, and it invites the natural objection that a tiny structure-sourced entropy cannot gate an order-unity dark-energy term — the deposited binding entropy is only $\epsilon_{\rm dep} \approx \Omega_{\rm m}\, f_{\rm coll}\, \langle\sigma_{\rm v}^2\rangle/c^2 \approx 1.5\times10^{-7}$ of the horizon entropy (the same six orders by which the binding *energy* falls short of $\rho_{\rm DE}$; §2). The objection is a category error, and the amplitude problem is *avoided*, not solved by amplification. The activation coordinate is not the deposited-entropy fraction $\Delta S_{\rm bind}/S_{\rm AH}$; it is the structure variance $x \equiv \sigma^2(z)/\sigma^2(0) = D^2(z)$, an order-unity quantity, which is the *argument* of the gate $f_{\rm sat}$ (the Press–Schechter collapsed fraction; §3.1, App. A.2). The deposited binding entropy enters in a *different* place: it fixes the entropy *weight* $\gamma \approx 1.5$ — the *shape* of the kernel — through the heat $\Delta S = E_{\rm bind}/T_{\rm AH}$ delivered to the horizon (§3.1, App. A.3). Hence $\Delta S_{\rm bind}/S_{\rm AH} \sim 10^{-7}$ does not imply $f_{\rm sat} \sim 10^{-7}$: they are different objects. With the *argument* (variance, O(1)) and the *weight* (entropy, $10^{-7}$) kept distinct, the gate activates simply because the structure variance grows by order unity — nothing is amplified and nothing is tuned. This is a consequence of the fixed inputs of §3, derived, not assumed.

The exact growth–expansion lock. Write the gate as $\phi \equiv f_{\rm sat}(x)$, x = D²(z), and define its derived response to the activation coordinate, $\chi_{\rm x} \equiv \partial\phi/\partial x = d\ f_{\rm sat}/dx = \gamma\ e^{(}-\gamma x)/(1-e^{(}-\gamma))$, which, once γ is fixed (§3.1), is an order-unity function with no free parameter — $\chi_{\rm x}(x=0) \approx 1.93$, $\chi_{\rm x}(x=1) \approx 0.43$. The fluid equation of state then follows exactly from continuity (the entropy-production identity above), using d ln φ/d ln $a = (\chi_{\rm x}/\phi) \cdot x \cdot (d$ ln x/d ln a) and d ln x/d ln a = 2g with g ≡ d ln D/d ln a: $1 + w_{\rm DE} = (1/3)[\,2\lambda\epsilon - (\chi_{\rm x}/\phi) \cdot 2x \cdot g\,]$, verified against the direct fluid EOS to $3 \times 10^{-3}$ (`run_chr_experiments.py`; suite tests V52–V53). Because $\chi_{\rm x}$, x, and g are each derived or directly measured, the phantom term is growth data, not a free function. This is the w(z) ↔ σ²(z) lock: given the measured expansion history, the dark-energy equation of state and the structure-growth history are not independent; inverting the relation predicts g(z) from w(z) to 0.6% within the model. A kinematic w(z) — including a CPL fit that reproduces SEDE's w(z) to within the DR3 precision — obeys no such relation. Expansion-only mimicry is therefore not enough: SEDE requires growth–expansion phase-locking. This is the test (P2 below) that separates the structure-sourcing mechanism from a curve fit — w from BAO/SNe/CMB distances versus g, D, fσ8 from RSD/weak lensing. It is parameter-free and uses nothing beyond §3; no criticality assumption enters.

The observable form of the lock (P2), and a GR consistency null. The lock derived above — the phantom part of w(z) equals $(1/3)(\chi_{\rm x}/\phi) \cdot 2x \cdot g$ reconstructed from growth data, obeyed by *no* kinematic dark energy — is the model's one parameter-free, degeneracy-breaking prediction. Its established observational realisation is the $E_{\rm G}$ estimator (the lensing-to-velocity amplitude ratio): SEDE is GR with smooth dark energy, so it predicts no gravitational slip, $E_{\rm G}(z) = \Omega_{\rm m}$,0/f(z), within <1% of ΛCDM and consistent with the DESI-DR1 + weak-lensing measurement to z ≈ 1 [arXiv:2507.16098] (`run_eg_test.py`). $E_{\rm G}$ is a *shared* GR null (with the GW-speed $c_{\rm T} = c$ and the zero-slip nulls SEDE passes), not a SEDE-vs-ΛCDM discriminator — at 0.3% it sits far below the ∼5–20% $E_{\rm G}$ errors; the model's only discriminator remains the deformation amplitude Δ (§5.6, §6). Failure of the lock (P2) would falsify the structure-sourcing mechanism itself — SEDE would then be an ordinary kinematic w(z) — so P2 is the test that separates the mechanism from a curve fit.

An optional near-critical layer (deferred to a companion paper). If, in addition, the horizon sits near the area↔volume spinodal — the flat-landscape limit, i.e. the $\Delta \to 1$ endpoint itself — the response sector becomes *near-critical* (the Critical Horizon Response regime) and acquires fluctuation-level signatures: a localised enhancement of large-scale activity at the transition z* ≈ 1.2 (P3); a small, nonzero, redshift-localised *separate-universe* (long-wavelength) modulation of growth/ISW peaked at z* (P4 — the kill test, distinct from the *broad, sub-horizon* signal that clustering dark energy with $c_{\rm s}^2 < 1$ would give, and from ΛCDM's exact zero); and a critical-slowing transient in $\gamma_{\rm growth}$ (P5). These are conditional, falsifiable predictions that add no fitted parameter but depend on the proximity-to-criticality hypothesis. Because they are not required for — and do not affect — the core background result, we develop them, with the separate-universe/super-sample-covariance formalism [arXiv:1711.07467] and the supporting figures, in a companion paper (in preparation); the corresponding kill test is within reach of Euclid/Rubin weak-lensing tomography.

Cross-horizon universality (speculative; companion). Whether Δ = 1 extends to black-hole horizons — giving $S_{\rm BH} \propto A^{3/2} \propto M^3$ and strongly altered primordial-black-hole evaporation — is a speculative extension logically separate from the core model, which requires the deformation only in the dark-energy sector. We do not adopt it: SEDE is consistent with the standard area law for astrophysical black holes, and the black-hole-vs-cosmic-horizon asymmetry is a genuine conceptual cost (§8.5), not a neutral prediction. We pursue this, and the Verlinde dark-matter connection (§7), in the companion paper.

# 7. Relation to other sectors

Dark matter. SEDE does not derive dark matter; it assumes the standard cold component. What it *does* provide are falsifiable *relationships*: the equation of state is locked to $\Omega_{\rm m}$ through the fixed-point background — the canonical model giving $w_0 \approx -1$ with a crossing set by the growth gate (§3.2; the closed-form $w_0 \approx -0.85$ enters only as a structural diagnostic, not the relation) — and the coincidence problem is reframed: dark energy's activation gate $f_{\rm sat}(z)$ tracks the (dark-matter-dominated) growth of structure, so "why now" corresponds to $f_{\rm sat}$ crossing ≈ 0.5 near z ≈ 1, shortly before matter–DE equality. These are testable links, not a unification, and we state them as such.

There is, however, a *principled* route to more, which we flag as a future direction and pursue in a companion paper, not here. The same volume-law dark-sector entropy SEDE uses for dark *energy* is the object from which Verlinde's emergent gravity (2016) derives apparent dark *matter* (baryonic displacement of the de Sitter volume-law entropy $\rightarrow$ the MOND scale $a_0 \sim cH_0$); if SEDE's Δ = 1 horizon entropy and Verlinde's volume-law de Sitter entropy are the same object, one volume-law postulate could in principle source both dark energy and apparent dark matter. We do not adopt this here: Verlinde's proposal is contested (it struggles with clusters and lensing and lacks a covariant CMB/structure completion — the regime SEDE *is* tested in), and our own cross-check (`run_verlinde_crosscheck.py`) finds only an *order-of-magnitude* consistency — the CKN/flatness normalisation sets $a_0$ within an O(1) factor of observed ($(c\sqrt{\Omega_{\mathrm{DE0}}}\,H_0)/2\pi \approx 0.73\,a_0$, $\sim$25% low), not a derivation. We record it as a motivated, partially-consistent target for the companion; SEDE's data fits use standard cold dark matter throughout.

Quantum and cross-field. SEDE's horizon is fractal/volume-law while its *bulk* is standard GR (holographic-DE scope), so it predicts a deformed horizon without a Lorentz-violating, foamy bulk. Two real datasets are consistent with this restricted-scope picture (they constrain bulk effects SEDE does not predict, so they do not contradict it): the Fermi GRB 090510 31 GeV photon excludes Planck-scale linear Lorentz violation ($E_{\mathrm{QG}} > several\ M_{\mathrm{P}}$), and the Fermilab Holometer's null result excludes bulk Planck-amplitude "holographic noise." The CKN reading gives a gravitating-vacuum cutoff $\Lambda_{\mathrm{UV}} \approx 4$ meV at the dark-energy scale (§8.2). We tested one cross-field extrapolation and it failed: the cosmological δ = 3/2 does *not* transfer to galaxy-cluster velocity statistics — a real SDSS/Coma fit gives a Gaussian (Tsallis q = 1.01, not the q ≈ 1.5 a naïve transfer would predict). We report this negative result plainly; the galaxy-dynamics link is the weakest cross-field claim and the data do not support it.

A contingent tension: cosmic birefringence. Minimal SEDE predicts β = 0 by symmetry (its parity-even thermodynamic scalar has no $\varphi F\tilde{F}$ coupling — the same structural fact as $\alpha_{\mathrm{T}} = 0$), whereas a nonzero isotropic rotation is reported (ACT DR6 0.215°±0.074°, 2.9σ [arXiv:2509.13654]). We treat this as a *real-but-contingent* threat, not a present falsification, and assess it in the §9 red-team (threat 2): the signal's cosmic, dark-energy-sourced nature is not established (miscalibration- degenerate α+β; the cleanest β×galaxy channel is null at SEDE's zero), but a confirmed cosmic β would favour a single-axion rival that does SEDE's crossing and more.

Where SEDE sits among dark-energy models — and what it borrows. SEDE is a thermodynamic model, but it is not outside the standard effective-field-theory description of dark energy, and locating it there removes its main field-theoretic disadvantage. In the EFT of dark energy (Gleyzes–Gubitosi–Piazza– Vernizzi), SEDE occupies the safe corner: the bulk is general relativity (holographic scope), so the Planck-mass run rate $\alpha_{\mathrm{M}} = 0$ ($\dot{G}/G$ = 0, §6/V7) and the tensor speed excess $\alpha_{\mathrm{T}} = 0$ — gravitational waves travel at c, so SEDE is GW170817-safe by construction ($|c_{\mathrm{T}}/c - 1| = 0$, against the $\sim 10^{-15}$ bound), having never switched on the higher Horndeski operators that GW170817 excludes. The dark energy is smooth with $c_{\mathrm{s}}^2 = 1$ (§3.4), so it is gradient-stable ($c_{\mathrm{s}}^2 > 0$) and does not cluster. SEDE is thus the ($\alpha_{\mathrm{T}} = \alpha_{\mathrm{M}} = 0$, $c_{\mathrm{s}}^2 = 1$) point — the *least*-constrained region of EFT-of-DE. Field-theory representation and stability through the crossing (`run_eft_lagrangian.py`): a *single* minimally-coupled scalar cannot reproduce the w = −1 crossing — for any P(X,φ), ρ + p = 2X $P_{\mathrm{X}}$, so the crossing forces $P_{\mathrm{X}} \rightarrow 0$ (verified: 1 + w → 0 at z ≈ 0.18), at which point the perturbation kinetic term $\alpha_{\mathrm{K}} \propto P_{\mathrm{X}} + 2X\,P_{\mathrm{XX}}$ passes through zero (a ghost) and $c_{\mathrm{s}}^2$ is ill-defined (the Vikman no-go). A *fundamental* ghost-free crossing nonetheless exists with braiding (Kinetic Gravity Braiding / Horndeski $G_3$(φ,X)□φ): the no-ghost determinant $Q_{\mathrm{S}} = \alpha_{\mathrm{K}} + (3/2)\alpha_{\mathrm{B}}^2$ stays positive even when $\alpha_{\mathrm{K}} \rightarrow 0$, because a nonzero $\alpha_{\mathrm{B}}$ carries it (we verify $Q_{\mathrm{S}} = 1.15 > 0$ at the crossing for a representative $\alpha_{\mathrm{B}} = 1.5\,\Omega_{\mathrm{DE}}$ realisation, with $c_{\mathrm{s}}^2 > 0$ throughout) [Deffayet et al. 2010; Kimura & Yamamoto 2010]. SEDE, however, needs neither: $\rho_{\mathrm{DE}} = T_{\mathrm{AH}} \cdot s_{\mathrm{grav}}$ is built from covariant horizon scalars, not a fundamental field, and is treated as an effective smooth fluid (PPF, $c_{\mathrm{s}}^2 = 1$) — gradient-stable ($c_{\mathrm{s}}^2 = 1 > 0$), with no propagating scalar to ghost, so the no-go simply does not apply (the thermodynamic content, not a Lagrangian, carries the prediction).

The structure-coupling — SEDE's most distinctive and least-confirmed ingredient (the structure-sourced-dark-energy idea itself traces to Gough's information dark energy, §7.6; what is specific here is the horizon-thermodynamic realisation) — is also free of the two instability classes that afflict coupled dark energy (`run_eft_stability.py`, V55). There is no gradient or ghost instability because $c_{\mathrm{s}}^2 = 1 > 0$; and there is no early-time large-scale instability because the interaction fraction $\Omega_{\mathrm{DE}}(z) \cdot |d \ln \varphi / d \ln a| \rightarrow 0$ at high z (it peaks at

$\approx$ 0.38 near z $\approx$ 0.5 and falls to $\sim 10^{-10}$ by recombination, since the dark-energy fraction itself vanishes as $f_{\rm sat} \to 0$). The coupling switches *off* in precisely the radiation/early-matter era where coupled-DE instabilities are seeded; the full perturbation evolution is validated end-to-end by CLASS-PPF (fσ8 to 0.1%, §4.4). So two of the apparent disadvantages relative to other programmes in fact *invert*: against Horndeski SEDE is less constrained (GW-safe), and against interacting dark energy it is more stable.

| framework | what SEDE shares or borrows | net position |
|---|---|---|
| ΛCDM (Planck) | empirical benchmark | honest limit: far less established — only DR3/Euclid closes this |
| quintessence (Ratra–Peebles) | dynamical DE; tracker/attractor crossing (§8.1, Result 12–13) | structure-locked w(z) with a non-tuned −1 crossing |
| CKN bound | the UV–IR energy bound fixes the DE scale (§8.2) | SEDE turns it into a cosmology; limit: needs the volume-law form on top |
| Li holographic DE | holographic DE with a horizon cutoff | uses the apparent horizon (no future-horizon causality/circularity); limit: volume-law is radical |
| standard Barrow-HDE (Saridakis 2020) | the Barrow entropy $S \propto A^{1+\Delta/2}$ | SEDE's Δ enters through the H-coupling λ = 1−Δ/2 *and* the structure gate, not the HDE density directly — so neighbouring Barrow-HDE fits that prefer $\Delta \ll 1$ [arXiv:2506.03019] constrain a *different* model and do not exclude SEDE's Δ = 1 (§9); limit: the volume-law endpoint is the radical input |
| Hu–Sawicki f(R) | viable late-time acceleration | keeps GR ($\alpha_{\rm M} = \alpha_{\rm T} = 0$): no screening/GW tension; limit: less mature |
| CPL | the data language; w(z) is CPL-degenerate in shape (§6) | adds the growth–expansion lock CPL lacks; limit: model-dependent, not a neutral fit |
| early dark energy | — | honest limit: does not address $H_0$ (out of scope, $r_{\rm d}$-pinned, §9) |
| Jacobson (1995) | Clausius $\to$ Friedmann; $\rho_{\rm DE} = T \cdot s$ (App. A.1) | applies horizon thermodynamics to DE; entropy-law change confined to the DE fluid (§8.3), not gravity |
| Horndeski / EFT-of-DE | the α-function language | sits at $\alpha_{\rm T} = \alpha_{\rm M} = 0$, $c_{\rm s}^2 = 1$ — GW170817-safe corner; less constrained, not less fundamental |
| Verlinde (2016) | volume-law de Sitter entropy; elastic/memory response (§6, §8.1) | independent motivation for the postulate; $a_0 \sim cH_0$ consistent at O(1) (§7) |
| interacting DE | the dark-sector energy-exchange ledger (CHR, §6) | coupling fixed to growth, $c_{\rm s}^2 = 1$, off at high z $\Longrightarrow$ instability-free |

The genuinely irreducible disadvantages, after these borrowings, reduce to two, and we state them plainly: SEDE is empirically younger than ΛCDM (the decisive tests are DR3/Euclid, §6), and it rests on the one volume-law postulate (§8.4) that motivation — Barrow, the GSL, CKN, Verlinde — supports but does not derive. Everything else in the table is either shared, borrowed, or an honest matter of scope.

Symmetry origin of the null predictions. SEDE's many "null" results are not independent assumptions: they follow from a *single* structural condition — *the dark sector adds no symmetry-breaking operator to the GR + Standard-Model effective action; it is a parity-even scalar* ($\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav} \cdot f_{\rm sat}$) *minimally coupled through* $T_\mu\nu$. Each precision null is then forced by a specific symmetry:

| null prediction | symmetry that forces it |
|---|---|
| β = 0 (no cosmic birefringence) | parity in electromagnetism → no φ F F̃ coupling |
| $\alpha_{\rm T} = 0$, $c_{\rm T} = c$ (GW170817-safe) | 2-derivative diffeomorphism-invariant gravity → no Chern–Simons / Galileon term |
| $\alpha_{\rm M} = 0$, Ġ/G = 0, no fifth force, equivalence principle | minimal (metric-only) coupling → no non-minimal $\xi\varphi^2 R$ |
| no Lorentz violation (Fermi GRB, Holometer) | bulk Lorentz invariance |

So β = 0 and $c_{\rm T} = c$ are *the same fact* (§7 birefringence; absence of a parity-violating / modified-propagation sector), and the whole cluster — the reason SEDE passes every GW, fifth-force, birefringence, and Lorentz null *at once* — is one principle, not a list. The de Sitter endpoints add a fifth: $f_{\rm sat} \to$ const (with H $\to$ const) $\Longrightarrow$ a maximally-symmetric de Sitter phase $\Longrightarrow$ w = −1 by symmetry (the dark-energy bracket of Result 12; App. A.7), with the crossing in between dynamical. We note the limit sharply: this symmetry argument protects SEDE's *safety*, not its *content*. The distinctive, load-bearing inputs — the volume law Δ = 1 (a geometric/thermodynamic postulate, §8.4), the coupling γ ≈ 1.5 (halo binding physics, §3.1), and the crossing redshift (a dynamical balance, Result 13) — are not symmetry-derivable; symmetry buys SEDE nothing there, and the one open postulate is untouched. The symmetry-protected cluster is precisely the part SEDE shares with any minimally-coupled smooth dark energy; what makes it SEDE remains thermodynamic and postulate-based.

## 7.6 Relation to information dark energy and other structure-sourced models

The idea that dark energy is *sourced by the entropy/information produced as structure forms* is not original to SEDE, and we state the priority plainly. It is due to M. P. Gough, in a sustained but under-cited programme that we date — from a NASA-ADS author search — to 2006–2008 ("On the Similarity of Information Energy to Dark Energy," 2006; "Information Equation of State," 2008), developed through *Holographic Dark Information Energy* [*Entropy* 13, 924 (2011), arXiv:1105.4461] to a recent observational analysis [*Entropy* 2025] that fits the stellar-mass-density history to the dark-energy equation of state from Planck/Pantheon+/DES/DESI, finding $\rho_{\rm DE} \propto SMD(a)^p$ with p ≈ 0.5 ($R^2$ = 0.93). In Gough's model dark energy is the Landauer energy of the information/entropy of star-heated gas and plasma. SEDE shares his central physics — structure's entropy *is* dark energy, with a self-regulating feedback (acceleration suppresses growth) that addresses the coincidence problem and predicts time-limited acceleration — and we credit it as the priority for the structure-sourced-dark-energy idea.

What distinguishes SEDE is the mechanism and a testable consequence, not the founding idea. (i) *Mechanism:* Gough uses the thermal/information entropy of hot baryons (∝ T, residing in the bulk); SEDE uses the gravitational binding entropy of bound structure deposited on the apparent horizon ($\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, Barrow volume-law Δ=1). These are physically different entropies in different places. (ii) *Prediction:* the two are not degenerate. SEDE's driver is the gravitational growth factor D(z) (dark matter), Gough's is the baryonic stellar-mass-density SMD(z) (star formation); computing both (`run` comparison, `output/fig_gough_vs_sede.png`) gives dark-energy histories that agree to ≲ 6% for z ≲ 0.6 — where supernova data are strongest, explaining why both fit current data — but diverge by up to ~50% by z = 3, SEDE staying flatter (more high-z dark energy) because gravitational growth persists where star formation declines. The coincidence that both land on a "0.5" exponent is superficial: Gough's is a power of stellar mass density, SEDE's is the H-exponent λ = 1−Δ/2 from Barrow Δ=1. (iii) SEDE adds a parameter-free dark sector and a w-crossing rather than a fitted CPL. So a clean way to separate them observationally is to ask whether dark energy tracks D(z) (gravity) or SMD(z) (stars), with the high-z dark-energy fraction (DESI Lyα, future surveys) as the lever.

Other structure-/entropy-linked programmes differ more sharply and are catalogued in `NOVELTY_CHECK.md`: entropic-force and Barrow/Tsallis holographic dark energy source the term from the horizon (structure downstream — and standard entropic-force DE in fact *struggles* with structure growth, which SEDE's design inverts); backreaction (Buchert, Räsänen) and timescape (Wiltshire) make acceleration an apparent inhomogeneity effect with no real dark energy; matter-creation/CCDM (Lima) replaces dark energy with a particle-creation pressure; and cosmologically-coupled black holes (Farrah et al.) use one special structure via vacuum interiors. A systematic priority pass over both INSPIRE-HEP and NASA ADS (citation-ranked, multiple phrasings; documented in `NOVELTY_CHECK.md`) returns no precedent for SEDE's specific construction — "dark energy = gravitational binding entropy of bound structure on the apparent horizon" (`abs:("dark energy" "binding energy" horizon)` yields zero relevant hits in either database) — nor for its terminology. So while the *general* structure-sourced-dark-energy idea belongs to Gough (since ~2006–2008), SEDE's *gravitational-horizon mechanism, its gravity-driven (D(z), not SMD) signature, and its parameter-free first-principles dark sector* appear to be new. We also ran Gough's model through our identical CAMB-marginalised pipeline (its $\rho_{\rm DE} \propto SMD^{0.5}$ expressed as a w(a) table): both are viable, with SEDE modestly preferred (ΔDIC $\approx -3.3$ vs Gough's $\approx -0.45$ relative to ΛCDM), and Gough's fit favouring a higher $H_0$ — consistent with his Hubble-tension claim. In summary, SEDE is a distinct, more-derived realisation of Gough's idea, decided from ΛCDM and from Gough alike only by the high-z dark-energy and growth data to come (§6).

# 8. The quantum-gravity underpinning

The energy-conservation account of §2–§3 took two horizon properties as given: the magnitude, $\rho_{\rm DE} \sim \rho_{\rm crit}$ rather than the QFT vacuum's $M_{\rm P}^4$, and the H-scaling λ = 1/2, which follows from the horizon entropy being volume-law ($S_{\rm grav} \propto V_{\rm AH}$; §3.3) with *no deformation parameter*. This section supplies their quantum-gravitational basis — it is the support beam under the construction, and a reader content to take the two scales as inputs may treat it as optional. We argue why the horizon entropy is volume-law rather than area-law, anchor the magnitude to the holographic bound and the flatness normalisation, and isolate the single statement that remains a postulate. Throughout, the Barrow parameter Δ appears only to *label* the area-to-volume interpolation ($S \propto A^{1+\Delta/2}$: Δ=0 area-law, Δ=1 $\Longleftrightarrow$ volume-law) and as the falsifiable test of the volume postulate (measured in §5.6, forecast in §6); it is not an input to the model.

## 8.1 Why volume-law, not area-law — consistency motivations (none a derivation)

The dark-sector entropy is the coarse-grained entanglement entropy contained in the apparent-horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R_{\rm AH}^3$ — equivalently the Barrow endpoint Δ=1 ($A^{1+\Delta/2} = A^{3/2} = R^3$). The open physical question is sharp: why the volume law, rather than the Bekenstein–Hawking *area* law? We give three considerations in the main line and collect three further, more speculative ones in Appendix G. None is a microscopic proof; together they motivate the volume endpoint and make the area-law branch disfavoured in the diagnostics.

1. Geometric ceiling caps the deformation (Result 9). The horizon's fractal dimension is $d_{\rm H} = 2 + \Delta$, capped at the space-filling value $d_{\rm H} = 3$ for a two-surface in three-space. Since (for a volume law in embedding dimension d) Δ = 2/(d−1), the case d = 3 gives Δ = 1. We are precise about what this delivers: the ceiling bounds the deformation, $\Delta \leq 1$ — it does *not* select Δ = 1. Landing *on* the endpoint requires, in addition, that the entropy is monotone in Δ (so the cap is saturated); since $\rho_{\rm DE} = T \cdot s$ makes the free energy F = E − TS vanish identically, entropy is the governing potential and the generalised second law is consistent with — is a fixed point of — the saturated (volume-law) state. We stress that this is a consistency check, not a selection principle: "total entropy does not decrease" is not "the parameter takes its entropy-maximising value," and the saturation of the cap *is* the volume-law postulate (§8.4) in effective form, not an independent derivation of it. Volume-law scaling is the signature of a thermalised, maximally-scrambled state (Page curve, eigenstate thermalisation).
2. The holographic bound fixes the scale, not the form. The CKN energy bound (§8.2) pins the *magnitude* but not the exponent — its own state count gives a *sub*-area-law (δ = 3/4), the opposite of an enhancement. The volume law is therefore a statement about the horizon's quantum *state* (typical / thermalised), which the energy bound cannot supply. This isolates the open content to one sentence: *the horizon is in a typical, volume-law-entangled state.*
3. The area-law branch is disfavoured directly by current data. *Saturating* the CKN bound — the natural "no-deformation" alternative — is algebraically the area-law case Δ = 0, whose equation of state ($w_0 = -0.49$)

and large early-dark-energy fraction are disfavoured in the conditional diagnostics of §5.6. On current data the volume branch is favoured over the area-law alternative, not merely an optional choice — though, as §5.6 stresses, the robust marginalised measurement is still to come.

Three further consistency motivations — a *driven non-equilibrium steady state* (the structure gate maintains the volume-law state despite a long scrambling time), a *roughening / Edwards–Wilkinson universality-class* analogy for the horizon surface, and the *independent volume-law entanglement entropy* of Verlinde's emergent-gravity programme — are collected in Appendix G. Each makes the postulate less *ad hoc*; none derives it.

## 8.2 Why $\rho_{\rm DE}$ ~ $\rho_{\rm crit}$ — the CKN bound and the vacuum-energy scale

The deformation fixes the *scaling* of dark energy with H but not its *magnitude*. The magnitude comes from the Cohen–Kaplan–Nelson (CKN) holographic energy bound (Principle 0): an effective field theory in a region of size L cannot hold more energy than a black hole of that size, ρ $L^3 \lesssim M_{\rm P}^2$ L, i.e. $\rho \lesssim M_{\rm P}^2/L^2$. With the IR cutoff at the cosmic horizon, L = c/H, this gives

$$\rho_{\rm DE} \lesssim M_P^2 H^2 \sim \rho_{\rm crit} \;\Rightarrow\; \Omega_{\rm DE} \sim \mathcal{O}(1).$$

The naïve QFT vacuum, ρ ~ $M_{\rm P}^4$, overshoots this by $\rho_{\rm P}/\rho_{\rm crit}$ ~ $10^{122}$ — the cosmological-constant problem. SEDE never invokes the QFT vacuum: dark energy is the horizon's conjugate thermal energy density, and the CKN relation supplies the horizon-scale upper bound $\rho_{\rm DE} \lesssim M_{\rm P}^2 H^2$. This is a bound, not an equality: the present amplitude is fixed by spatial flatness ($\Omega_{\rm DE0} = 1 - \Omega_{\rm m}$), in the same bookkeeping sense that ΛCDM fixes $\Omega_\Lambda 0$ — once the volume-law branch is chosen, the bound is saturated to O(1) by that normalisation today. We note explicitly that the saturation is a present-day statement: since $\rho_{\rm DE} \propto H$ (volume law) while the bound ∝ $H^2$, the ratio $\rho_{\rm DE}/(M_{\rm P}^2 H^2) = \Omega_{\rm DE}(z)$ falls at higher z (0.70 → 0.22 → 0.03 at z = 0, 1, 3), so dark energy sits *below* the CKN bound in the past — consistent, as CKN is an upper bound. So CKN *addresses* the vacuum-energy-scale problem (it explains why the natural gravitational horizon scale is $M_{\rm P}^2 H^2$ ~ $\rho_{\rm crit}$, not $M_{\rm P}^4$) rather than deriving the magnitude from first principles. The same bound, read as a UV–IR relation, gives a gravitating-vacuum cutoff $\Lambda_{\rm UV} = (M_{\rm P}\, H_0)^{1/2} \approx 4$ meV — numerically the observed dark-energy scale (§7).

## 8.3 The seam, BBN-safety, and the holographic scope

A naïve *microscopic* volume-law entropy density would be Planckian, so a naïve $\rho_{\rm DE} = T \cdot (S_{\rm vol}/V)$ would overshoot $\rho_{\rm crit}$ by $(M_{\rm P}/H)^\Delta$ ~ $10^{61}$ — the square root of the $10^{122}$ hierarchy (Result 11). The *same* factor controls three apparently separate issues — this seam, the BBN bound on a Barrow-modified Friedmann equation, and the modified-gravity-vs-holographic fork — which are one object. The resolution is the holographic-DE scope adopted in §2: the deformation acts only on the dark-energy fluid (magnitude from CKN, scaling from Barrow), not on the gravitational sector. The early-universe expansion is then exactly standard ($\Delta N_{\rm eff} = \Delta Y_{\rm p} = 0$, verified; BBN-safe), and the naïve T·(S/V) is precisely the modified-gravity path the scope forbids.

This is the load-bearing assumption of the model. In the *modified-gravity* reading — Barrow entropy deforming the Friedmann equations [arXiv:2110.00059] — BBN forces Δ $\lesssim$ 1.4×$10^{-4}$ [Barrow, Basilakos & Saridakis, arXiv:2010.00986] and SN+BAO+H(z) give Δ ~ $10^{-4}$ (the area law at 2σ [arXiv:2108.10998]), which read literally would exclude SEDE's Δ = 1. Those bounds do *not* apply to SEDE: it keeps standard gravity and lets the volume-law entropy act only on the dark-energy fluid via $\rho_{\rm DE} = T \cdot s$ (§2, App. A.1), so there is no extra Friedmann term, no BBN deviation, and Δ is unconstrained by them. The headline Δ = 1 thus rests on this single scope choice — legitimate and standard (it *is* the holographic-dark-energy framework), but not free: a reader who rejects it recovers the tightly-bounded modified-gravity Barrow cosmology instead.

## 8.4 The one irreducible postulate

Postulate (volume-law horizon). The cosmic apparent horizon carries the coarse-grained *entanglement* entropy of its volume, $S_{\rm grav} \propto V_{\rm AH} \propto R^3$ — a constant horizon entropy density, i.e. a thermalised, volume-law (non-extensive) state — rather than the Bekenstein–Hawking area law.

The results in §8.1–§8.3 are bounds, consistency arguments, or motivations *given* this postulate; the postulate itself is not derived from a microscopic theory of quantum gravity. §8.1 *reduces* it (the CKN bound fixes the scale, so the only open content is the horizon's entanglement *state*), *motivates* it (non-extensivity, the GSL, the driven steady state, the Edwards–Wilkinson universality class, and the independent volume-law entanglement entropy of Verlinde's emergent gravity), and shows the *area-law branch is disfavoured by current diagnostics* — but a microscopic state count that delivers the volume law remains the model's single open foundational item (§9). Two points of emphasis. First, the model carries no fitted deformation parameter: Δ is not an input but the falsifiable *test* of this postulate — current data already prefer the volume endpoint and disfavour the area-law branch (§5.6, a conditional diagnostic), with the robust marginalised measurement to come from DESI DR3 + Euclid (§6), alongside cross-horizon predictions (a deformed black-hole entropy $S \propto A^{3/2}$; §7, presented as speculative). Second, the open question is now stated plainly — *why is the horizon entropy volume-law (coarse-grained entanglement) rather than area-law?* — rather than hidden in a value of a tunable parameter. We do not claim to have derived quantum gravity; we claim to have reduced its dark-energy content to one testable statement and removed it as a fitted parameter.

On the plausibility of a volume law (genre, not derivation). A volume-law horizon entropy is a strong hypothesis, but volume-law *entanglement* entropy is at least not exotic in cosmological field theory. The entanglement entropy of a field subregion — area-law in the de Sitter adiabatic vacuum — grows and saturates to a *volume law* after a de Sitter→flat transition, ending in a partially thermalised (generalised-Gibbs) state [Khlebnikov & Sheoran, arXiv:1907.00487]; volume-law entanglement also arises from field–curvature coupling [arXiv:2306.08357]. We cite these only to establish *genre and plausibility* — that volume-law entropy is a real feature of cosmological QFT — and we explicitly resist over-reading them. Two caveats: (i) these results concern the entanglement entropy of field *subregions* under specific transitions, not the cosmic-*horizon* entropy SEDE invokes; and (ii) their assignment (de Sitter = area-law, flat = volume-law) is *not* SEDE's and in fact runs opposite to it (SEDE places the volume law in the de Sitter brackets). So the literature makes a volume-law hypothesis less exotic, but it neither *anticipates* nor *derives* the specific postulate — which remains the model's one open foundational item (§9). (We note for completeness that Padmanabhan's holographic-equipartition route, $dV/dt = N_{\rm sur} - N_{\rm bulk}$, does *not* help here: its surface count is area-law, $N_{\rm sur} = A/\ell_{\rm P}^2$, and it reproduces the *standard* area-law Friedmann equation — it is the baseline SEDE departs from, not independent support for the volume law.)

## 8.5 The postulate is a dof-*counting* claim, not a thermalisation claim

The phrase "volume-law" conflates two claims of very different status. (i) The state: are the horizon degrees of freedom maximally entangled (thermal) or in a low-entanglement ground state? (ii) The counting: does the *number* of horizon dof grow as the area, $N \propto R^{d-2}$, or the spatial volume, $N \propto R^{d-1}$? Since S ∼ (state factor) × N, the deformation Δ — through $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$ — is fixed by the counting (area→Δ=0, volume→Δ=1 in d=4); the state only sets whether the prefactor is maximal. SEDE's postulate is, precisely, the counting claim.

*The state half is settled, and is not peculiar to SEDE.* Maximal scrambling has a precise meaning in the Sachdev–Ye–Kitaev model [Sachdev & Ye 1993; Kitaev 2015], the canonical chaotic, black-hole-dual system. But maximal scrambling does not fix Δ: a black hole is a *maximal* scrambler yet area-law (S = A/4, Δ=0, reproduced by string microstate counts). Scrambling is therefore shared with the Δ=0 black hole; and SYK, being all-to-all (geometry-free), cannot even pose the counting question. (An explicit Majorana-SYK diagonalisation confirming the state is maximally entangled — Wigner–Dyson statistics, Page-value saturation, complete OTOC scrambling — is given in Appendix G; it settles the *state*, not the counting.)

*The counting half is where SEDE lives, and area is the default.* In causal-set quantum gravity — the setting where the counting question is natural — both scalings appear, but the *canonical* horizon-entropy object, the causal-link count that recovers the Bekenstein–Hawking law [Sorkin and collaborators], is the area one ($R^{d-2}$), in flat space and identically in de Sitter. So even in the framework most hospitable to volume-counting, area-law is the default, and SEDE's Δ=1 is the specific, non-default identification of horizon entropy with the bulk count. (The causal-set sprinkle measurements, the entropy-bound check, and a Ryu–Takayanagi holographic-code realisation — volume-law requires *nonlocal* horizon connectivity — are in Appendix G; their verdict is that the volume count is allowed and realisable but not selected: no consistency argument available to us derives it, and none excludes it.)

The sharpened postulate, properly located. The one open item is therefore *not* about thermalisation or chaos (settled, and shared with area-law black holes) but is purely a dof-counting statement:

> the cosmic horizon's entropy counts its bulk (volume-scaling, $N \propto R^3 \Longleftrightarrow d_H = 3$) degrees of freedom rather than its boundary (area-scaling, links) ones — a count that exceeds the holographic area bound by $R/\ell_P$ (the seam of §8.3).

It reduces to two deciders. *Theoretically* it is precisely *"is the Hilbert-space dimension of the de Sitter static patch $e^{Area/4}$ or $e^{Volume}$?"* — a frontier problem a complete dS holography would settle. *Empirically* it is the deformation Δ: Δ=0 (area) is already disfavoured by present data (§5.6), and DESI DR3 + Euclid will measure Δ to ~0.09 (§6). SEDE's postulate is thus not an arbitrary assumption but a *consistent, realisable bet on a well-posed open question of quantum gravity*, with a decisive test imminent. Tellingly, the one horizon we can check today — the black hole — is area-law; we name that asymmetry as a genuine conceptual cost (§7), not a neutral prediction.

# 9. Discussion: open problems & honest assessment

The one open item. Every dark-sector input is fixed by a stated choice (§3) given the single postulate that the horizon is volume-law-entangled (§8.4). That postulate is motivated by gravity's non-extensivity and the second law but is not derived from a microscopic state count; it is the model's one irreducible input. We have argued it is not closable by reframing (canonical vs microcanonical accounting gives either area-law $\lambda = 1$ or Planckian-volume $\lambda = 1/2$, never both for free) — it is a genuine open problem, and a *falsifiable* one (§6).

Scope and tensions. $H_0$ is out of SEDE's scope (it is $r_d$-pinned; the model does not address the distance-ladder tension and we do not claim it does). The structure-sourcing *mechanism* itself — the defining SEDE claim that $f_{sat}$ tracks growth — is only weakly confirmed: a geometry-vs-structure consistency test (E1; the null test $\rho_{DE} = G[D(z)]$ with $f_{sat}$ read independently off the *expansion*, DESI $D_H/r_d \to H(z)$, and off *structure growth*, RSD $f\sigma 8 \to \sigma 8(z)$) tracks at $z < 1$ but is RSD-limited at $z > 1$ and remains inconclusive (current data: $\chi^2$/dof = 0.40 in the RSD-error budget, 1.40 in the precise-geometry budget; Figure 9). The pipeline is implemented and staged (`run_e1_mechanism.py`) — it runs now and is ready to apply the moment Euclid/LSST weak-lensing tomography sharpens the structure leg, at which point the forecast of §6 makes it decisive ($\sim 4\sigma$). Its data vector and CONFIRM/REFUTE thresholds are pre-registered (PREREGISTRATION.md §F) so the staged test cannot be tuned after the data arrive. We have separated this mechanism into a derived, parameter-free part — the growth–expansion lock P2, which follows from $f_{sat} = f(D^2)$ with no extra assumption (§6), testable with exactly that data — and an optional near-critical layer (deferred to a companion paper) that supplies a further kill test. But a *derived consistency relation* is not yet a *confirmation*, and the mechanism stays unconfirmed until the lock P2 is measured. We stress this is a *near-term* gap, not a permanent one: a Fisher forecast (§6, `run_mechanism_forecast.py`) shows the growth-side mechanism test reaches $\sim 4\sigma$ with combined Euclid/LSST weak lensing + CMB-lensing + clusters + RSD — so the distinctive claim is decidable on the same timescale as the amplitude Δ, not deferred indefinitely. And the cosmic- birefringence signal (§7) is a live tension. We emphasise the honest framing: SEDE's *phenomenology* (w crossing −1, the volume-law endpoint) is mildly favoured in the current compiled-data analysis with an encouraging out-of-sample check, but its *distinctive mechanism* (the $f_{sat}$ gate tracking structure growth) is the unconfirmed frontier.

Threats from the recent literature (a red-team). We list the strongest recent results that could damage SEDE, with our honest assessment of each.

1. *Holographic DE disfavoured by full CMB / the early–late tension.* Wu et al. [arXiv:2509.02945] find that holographic and Ricci DE are disfavoured by full ACT DR6 + DESI through an early-vs-late tension that compressed (R, ℓ_A) priors hide — a direct challenge, since SEDE's headline ΔDIC used compressed CMB. We tested it. Marginalised full primary likelihood: a CAMB-in-the-loop fit on the full Planck $plik_{lite}$ TTTEEE + lowℓ TT/EE, with the five cosmological parameters and the Planck nuisance *all free* (not θ*-fixed), gives $\Delta\chi^2$(SEDE−ΛCDM) = −0.43 (`run_full_cmb_mcmc.py`), the preference sitting in the high-ℓ peak structure (−0.37) — *negative*, i.e. mildly favouring SEDE, not the large *positive* $\Delta\chi^2$ the HDE early–late tension would produce; the best-fits are physical ($\Omega_m = 0.295/0.315$ for SEDE/ΛCDM). This is corroborated by two θ*-matched single-point proxies — the same $plik_{lite}$ ($\Delta\chi^2_{CMB} = -0.50$, `run_plik_check.py`) and the official ACT DR6 CMB-lensing likelihood ($\Delta\chi^2_{ACT}$-lens = −0.32, `run_act_lensing.py`, validated against

the package's fiducial $\chi^2 = 14.06$). ACT DR6 lensing re-evaluated at the marginalised primary best-fit (not a fixed fiducial) gives $\Delta\chi^2_{\rm ACT}$-lens = −0.31 (ΛCDM 14.10 → SEDE 13.79), and the low-$\ell$ ISW-sensitive channel is flat (−0.03). All are $|\Delta\chi^2| < 1$: SEDE does not inherit the standard-HDE tension, because it has *less* early dark energy than ΛCDM ($f_{\rm sat} \to 0$, $\Omega_{\rm DE}(z=3) \approx 0.035 < 0.040$), not the runaway high-z behaviour that breaks original HDE. Full non-lite plik (foregrounds sampled): we have since run the *complete* high-$\ell$ plik TTTEEE likelihood with all 15 foreground/calibration nuisance parameters sampled at their Planck priors (`run_full_cmb_mcmc.py --fullplik`), the gold-standard primary CMB — it gives $\Delta\chi^2$(SEDE−ΛCDM) = −0.33 (TTTEEE −0.27, lowl.TT −0.23, lowl.EE +0.17), confirming the foreground-marginalised $plik_{\rm lite}$ result (−0.43) to within the minimiser noise. So letting the foregrounds float does not reveal a hidden tension — the standard-HDE early–late wall is simply absent for SEDE. *Residual scope:* only the ACT DR6 *primary* multifrequency likelihood remains not run (ACT *lensing*, the DE-sensitive channel, is; §above) — formally deferred under the pre-registered trigger (PREREGISTRATION.md §E; `act_dr6_mflike` + multi-GB products not installed), with the falsifiable committed expectation $|\Delta\chi^2| < 1$ given three concordant CMB datasets already at that level; a full MCMC for ΔDIC is deferred (expected ≈ $\Delta\chi^2$, $p_{\rm D}$ shared). The full primary likelihood is also folded *into the joint* (§5.1).

2. *Cosmic birefringence — a contingent threat, with a rival waiting if it is confirmed.* SEDE predicts $\beta = 0$; a nonzero *rotation* is reported (ACT DR6 2.9σ [arXiv:2509.13654]; Planck ~0.3°), and *if* it is cosmic and dark-energy-sourced, a single axion explains β + acceleration [arXiv:2504.17638] and an interacting axion sector reproduces the phantom crossing + β + dark matter [arXiv:2506.12589] — a parsimony threat, since the rival does SEDE's crossing and more with one field. But the cosmic, DE- sourced nature is *not established*: the measurement is of α + β (miscalibration-degenerate), Planck and ACT disagree by >3σ, SPIDER is incompatible with the coupling [arXiv:2510.25489], and the β × LSS cross-correlation is null (p ≈ 37%, [arXiv:2509.22273]) — at SEDE's predicted zero, not the axion (§7). So this is a *contingent* threat, not a present falsification, and the cleanest current channels favour β = 0. The discriminator is the pair {w-crossing, β}; β = 0 is itself symmetry-derived (§7), tied to GW-safety, so a confirmed cosmic β would also predict GW parity violation.
3. *The DDE signal may be a low-z SN systematic.* Huang, Cai & Wang [arXiv:2502.04212] show the DESI evolving-DE preference is biased by an external low-z SN sample in DESY5 and drops to <2σ once corrected. *Defence (partial):* the bias is DESY5-specific ("in contrast to the uniform behaviour of PantheonPlus"), and SEDE's headline uses Pantheon+; more decisively, SEDE's leave-one-probe-out drops supernovae entirely and stays preferred ($\Delta\chi^2 = -2.47$), and $F_{\rm AP}/tracer$-LOO are SN-free. SEDE is not SN-driven — but if the field's evolving-DE evidence softens, SEDE's *context* erodes, and its "suggestive, not established" framing must hold.
4. *The nearest Barrow-HDE fit prefers $\Delta \ll 1$.* Luciano, Paliathanasis & Saridakis [arXiv:2506.03019] fit standard Barrow-HDE to DESI DR2 + SN + CC and obtain Δ < 0.47–0.54 (Δ = 0 within 1σ, mild positive tendency) — which, read literally, disfavours SEDE's Δ = 1. *Defence (shown, not asserted; Fig. 10, `run_delta_orthogonality.py`):* both models share the scaling $\rho_{\rm DE} \propto H^{2-\Delta}$; the *only* difference is SEDE's structure gate $f_{\rm sat}(z)$, and the gate is exactly what decouples Δ from the observable the Barrow-HDE bound rests on. At the same Δ = 1 the two models differ sharply: SEDE's gate ($f_{\rm sat} \to 0$ at high z) holds the early-dark-energy fraction to $\Omega_{\rm DE}(z=3) = 0.029$ — CMB-safe — whereas un-gated Barrow-HDE has $\Omega_{\rm DE}(z=3) = 0.147$ (5× larger) and $w_0 = -0.77$ with no w = −1 crossing, versus SEDE's $w_0 = -1.0$ in DESI's crossing quadrant. So the Δ-constraint derived from *Barrow-HDE's* early-DE/EOS does not transfer: SEDE's Δ = 1 produces viable observables precisely where Barrow-HDE's Δ = 1 does not, and in SEDE's own construction Δ = 1 is data-preferred over Δ = 0 (§5.6). The bound constrains a sibling model, not SEDE — and Fig. 10 shows the orthogonality rather than asserting it.

Figure 10. Why the standard Barrow-HDE bound $\Delta \ll 1$ does not constrain SEDE's Δ = 1. Both models share the holographic scaling $\rho_{\rm DE} \propto H^{2-\Delta}$; the only difference is SEDE's structure gate $f_{\rm sat}(z)$. *Left:* the early-dark-energy fraction $\Omega_{\rm DE}(z=3)$ — the quantity the CMB and the standard-HDE EOS fit actually constrain — versus Δ. Un-gated Barrow-HDE (red) carries large, strongly Δ-dependent early DE ($\Omega_{\rm DE}(z=3) = 0.15$ even at Δ = 1, rising to 0.70 at Δ = 0), so its fits pin Δ < 0.5; SEDE's gate (blue) drives $f_{\rm sat} \to 0$ at high z, holding $\Omega_{\rm DE}(z=3) \approx 0.03$ — at or below the ΛCDM/CMB-safe level (grey dashed) — for *all* Δ. At Δ = 1 the two models differ five-fold (0.029 vs 0.147), and SEDE additionally crosses w = −1 ($w_0 = -1.0$) where un-gated Barrow-HDE does not ($w_0 = -0.77$). *Right:* consequently the data prefer orthogonal Δ in the two parametrisations — $\hat{\Delta} < 0.5$ for Barrow-HDE

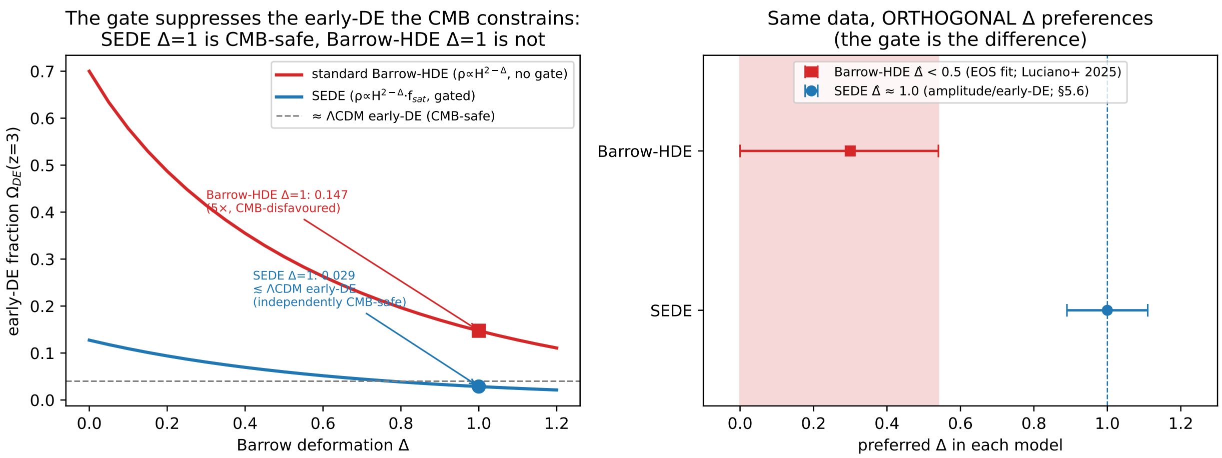


Figure 1: Fig10

(EOS/early-DE; Luciano+ 2025) and $\hat{\Delta} \approx 1.0$ for SEDE (amplitude/early-DE; §5.6) — with no contradiction. The gate is the difference. (`run_delta_orthogonality.py`.)

Net: the two threats that targeted the model itself (full-CMB tension; the Barrow-Δ bound) are answered by direct tests or a stated model distinction; the two that target the *context* (birefringence; the low-z SN systematic) are real and partially outside SEDE's control — birefringence especially.

Evidential status. SEDE is a falsifiable, fixed-parameter, Λ-free alternative to ΛCDM, mildly preferred (suggestive ~2–3$\sigma$) with an encouraging out-of-sample check, resting on one open postulate and a few stated theoretical choices. It is stronger than "indistinguishable from ΛCDM" and weaker than "established." Its value is twofold: a *physical story* (no fine-tuned Λ, w(z) from $\Omega_\mathrm{m}$, the coincidence reframed) and a *decidable prediction* (the volume law, $\Delta = 1$). The data, not the theory, will settle it.

## 10. Conclusions

We have presented Structural Entropy Dark Energy: a fixed-parameter dark-energy model in which dark energy is identified with the conjugate thermal energy density of a volume-law-entangled cosmic horizon and activated through a structure-growth gate. The model is a single Friedmann fixed-point equation with no fitted dark-sector parameter — λ, w(z), γ, and $c_s^2$ are each fixed by a stated theoretical choice rather than tuned — that replaces the cosmological constant by a structure-gated, horizon-coupled term. It addresses the vacuum-energy-scale problem (the scale is the CKN holographic bound, normalised by flatness, not the QFT vacuum) and reframes the coincidence problem (the activation gate tracks the late-time maturation of the structure-growth history). In a marginalised, CAMB-in-the-loop joint analysis it is mildly preferred over ΛCDM (ΔDIC $\approx -2.9$); the preference is suggestive (calibration ~2–3$\sigma$, not obviously flexibility) with an encouraging pre-DESI $\to$ DESI out-of-sample check, and its equation of state crosses $-1$ in DESI's evolving-DE quadrant. Its quantum-gravitational content reduces to one falsifiable postulate — a volume-law horizon entropy. The sharp future test is near: under the stated Fisher assumptions, DESI DR3 and Euclid could constrain the effective deformation Δ to $\sigma \approx 0.09$, separating SEDE's volume-law horizon from a smooth one at ~11$\sigma$. SEDE is suggestive, not established — and, crucially, it is decidable.

## Code and data availability

All code, data vectors, and analysis pipelines are publicly available at `github.com/spsingularity/SEDE`. A single entry point, `reproduce_all.py`, regenerates every number and figure in this paper from staged drivers; the model and likelihood live in the `sede/` package, and the verification suite (`sede/verification.py`) checks the quoted results numerically. The observational inputs are standard public datasets (DESI DR2, Pantheon+/DES-5YR/Union3, Moresco cosmic chronometers, Gold-2018 fσ8, compressed Planck, eBOSS DR16), retrieved via the documented loaders. A tagged release with an archival DOI will accompany journal submission.

## Use of AI tools

Artificial-intelligence tools (large language models) were used to assist with drafting and editing the manuscript, with developing and cross-checking the analysis code, and with literature searches. The author conceived the model, directed and independently verified all analyses, and takes full responsibility for the content, including all derivations, results, and conclusions. No AI tool is an author.

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# Appendix A — Key results, prescriptions, and assumptions

This appendix documents the six load-bearing items, labelled by their actual epistemic status — one defining ansatz (A.1), supporting derivations (A.2, A.4, A.5), a fixed prescription (A.3), and a motivated postulate (A.6). The headings give that status; the parenthetical "Result/Prescription/ Motivation N" numbering corresponds to the repository code labels `THEOREM_N` in `sede/theory.py` (retained there for code stability, *not* a claim that each is a mathematical theorem). We work in units $c = \hbar = k_{\mathrm{B}} = 1$ and write $8\pi G = 1/M_{\mathrm{P}}^2$. Source statements `THEOREM_1`, `THEOREM_3`, `THEOREM_4C`, `THEOREM_5`, `THEOREM_8`, `THEOREM_9`; the remaining items (2, 4, 4B, 6, 6B, 10–13, 5D) are catalogued at the end with one-line statements.

## A.1 The conjugate horizon-fluid ansatz: $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$

Claim. SEDE *identifies* the dark-energy density with the conjugate energy of the horizon entropy, $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav} \cdot f_{\rm sat}$, with $s_{\rm grav}$ the (volume-law) horizon entropy density and $f_{\rm sat} \in [0,1]$. The Clausius relation supplies the *dynamics*; the absolute density ρ = T·s is the model's defining ansatz, motivated by horizon thermodynamics — not a theorem.

Setup. At the flat-FRW apparent horizon $R_{\rm AH} = 1/H$, the dynamical Cai–Kim temperature is $T_{\rm AH} = (H/2\pi)(1-\epsilon/2)$, $\varepsilon \equiv -\dot{H}/H^2$. We adopt the Gibbons–Hawking limit $T_{\rm AH} = H/2\pi$ ($\varepsilon \to 0$), the leading temperature appropriate for the de Sitter-attractor dark-energy sector; the dynamical (1 − ε/2) correction is sub-leading where dark energy matters and is disfavoured in full (§2, §6), so it does not enter the calibrated E(z). The gravitational entropy is the coarse-grained entanglement entropy in the horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R_{\rm AH}^3$ (§3.3, §8.1), i.e. a *constant* entropy density $s_{\rm grav} = s_0$ (in contrast to the Bekenstein–Hawking area-law density $s_{\rm AH}$=¾$M_{\rm P}^2 H \propto H$, which would instead give the area-law model $\rho_{\rm DE} \propto H^2$ — the branch disfavoured in the conditional diagnostics of §5.6).

Construction. *Step 1 (Clausius — the rigorous, dynamical content).* Hayward's first law for a dynamical horizon, $\delta E = T_{\rm AH}\, dS_{\rm AH} + W$ δV with work density W = (ρ − p)/2, applied to a shell of matter crossing the horizon in time dt with flux −dE = (ρ + p)H V dt, gives the Clausius relation (ρ + p)H $V = T_{\rm AH}$ Ṡ_AH. This relates *changes* and is standard horizon thermodynamics.

*Step 2 (Bousso bound as motivation for a bounded* $f_{\rm sat}$*).* The covariant entropy bound gives $S_{\rm matter} \leq A/(4l_{\rm P}^2) = S_{\rm AH}$ on any light-sheet bounded by the horizon. This *motivates* interpreting $f_{\rm sat}$ as a bounded effective saturation fraction; in SEDE we impose $0 \leq f_{\rm sat} \leq 1$ and $f_{\rm sat}(0) = 1$ with the redshift dependence supplied by the halo prescription (A.2/A.3), and we do not identify $f_{\rm sat}$ literally with $S_{\rm matter}/S_{\rm AH}$ (consistent with the small binding-entropy budget, App. F).

*Step 3 (the SEDE identification — the ansatz).* SEDE posits the dark-energy density to be the conjugate energy of the volume-law horizon entropy,

$$\rho_{\rm DE} \equiv T_{AH}\, s_{\rm grav}\, f_{\rm sat} = \frac{H}{2\pi}\, s_0\, f_{\rm sat} \;\propto\; H\, f_{\rm sat}.$$

Fixing the one constant $s_0$ by spatial flatness [$\rho_{\rm DE}(0) = \Omega_{\rm DE0}\, \rho_{\rm crit}$,0, $f_{\rm sat}(0) = 1$] gives

$$\rho_{\rm DE}(z) = \Omega_{\rm DE0}\, \rho_{\rm crit,0}\, \frac{H}{H_0}\, f_{\rm sat}(z) = \Omega_{\rm DE0}\, \rho_{\rm crit,0}\, E(z)\, f_{\rm sat}(z),$$

i.e. the fixed-point term $\Omega_{\rm DE0} \cdot E \cdot f_{\rm sat}$ of eq. (1) — the volume-law (λ = 1/2) scaling, *not* the area-law H². This is an identification motivated by horizon thermodynamics, not a derivation: the Clausius relation (Step 1) fixes the dynamics, while the absolute density is the postulate (§8.4).

*Step 4 (scale / coincidence).* $\rho_{\rm DE}$ ~ $M_{\rm P}^2 H_0^2$ ~ $(10^{-3}\ \text{eV})^4$ is small because $H_0/M_{\rm P}$ ~ $10^{-61}$. The CKN bound (§8.2) *explains* why this horizon scale is ~$\rho_{\rm crit}$ — as an upper bound, taken here to be saturated to O(1) by the flatness normalisation above — replacing the QFT-vacuum conflation behind the $10^{120}$ problem rather than deriving the magnitude from scratch. ■

## A.2 Result 3 — the structure gate $f_{\rm sat}(z)$

Claim. $f_{\rm sat}(z) = [1 - e^{-\gamma x(z)}]/[1 - e^{-\gamma}]$, where x ≡ σ8²(z)/σ8²(0) = D²(z) (D the linear growth factor, D(0) = 1).

Setup. We model the cumulative activation gate directly (dimensionless, avoiding any mix of area- and volume-law densities) through a normalised growth-weighted deposition kernel K(t) ∝ (dσ8²/dt) $e^{-\gamma\sigma 8^2/\sigma 8^2(0)}$ — the structure-growth rate dσ8²/dt > 0 times a Boltzmann factor $e^{-\gamma x}$ for the exponential rarity of massive halos (γ from A.3) — taking $f_{\rm sat}$ to be its normalised cumulative from early times to epoch z (cosmic time t),

$$f_{\rm sat}(z) = \frac{\int_0^{t(z)} K\, dt}{\int_0^{t_0} K\, dt},$$

so that $f_{\rm sat}(\infty) = 0$ (early, t→0) and $f_{\rm sat}(0) = 1$ (today, $t = t_0$) by construction (the closed-form gate below is this cumulative-kernel form, not a relaxation ODE).

Proof. The kernel integrated over cosmic time is K dt = (dσ8²/dt) $e^{-\gamma\sigma8^2/\sigma8^2(0)}$ dt = $e^{-\gamma u}$ dσ8² with u ≡ σ8²/σ8²(0); the Hubble factors cancel and the integral reduces to one over u. With u running from 0 (z′→∞, no early structure) to x(z) ≡ σ8²(z)/σ8²(0) = D²(z),

$$f_{\rm sat}(z) = \frac{\int_0^{x(z)} e^{-\gamma u}\, du}{\int_0^1 e^{-\gamma u}\, du} = \frac{1 - e^{-\gamma x(z)}}{1 - e^{-\gamma}}, \qquad x(z) = D^2(z),$$

where the denominator (x = 1, i.e. z = 0) enforces $f_{\rm sat}(0) = 1$. Checks: $f_{\rm sat}(0) = 1$; $f_{\rm sat}(z \gg 1) \to 0$ as x → 0; $f_{\rm sat} \in$ [0,1] for x ∈ [0,1]. The form is the CDF of a truncated exponential of rate γ on x ∈ [0,1] — the normalised activation fraction reached by epoch z. ■

## A.3 Prescription 4C — the fixed structure coupling γ ≈ 1.496

Claim. The entropy weight exponent is p = 5/3, giving γ = d ln $\Sigma_{\rm S}^{(5/3)}/d$ ln σ8² = 1.496 (the derivative is w.r.t. the variance σ8², the gate argument x = D² = σ8²/σ8²(0) of A.2; equivalently ½ d ln $\Sigma_{\rm S}/d$ ln σ8 = 1.496, i.e. d ln $\Sigma_{\rm S}/d$ ln σ8 = 2.993). A self-similar reduction gives the closed form γ = (p−1)⟨1/α⟩_Σ with α ≡ −d ln σ/d ln M the effective spectral slope over the entropy-weighted mass range; it reproduces the exact mass-function integral to 0.1% (`run_gamma_systematics.py`, `BINDING_ENERGY_MECHANISM.md` §7). Entropy is deposited by the $\nu_{\rm S} \approx 2.4$ (~2σ) clusters at $M_{\rm S}$ ~ 10^14.5 M⊙/h, giving an independent cluster-abundance handle; p = 1 gives ≈ 0.27 (∼5× below p = 5/3), a weak gate versus the real one.

Proof. In the SEDE horizon-fluid ansatz (A.1), $f_{\rm sat}$ is the effective fraction of the cosmic-horizon entropy capacity that is gravitationally activated, and $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav} \cdot f_{\rm sat}$ — so the entropy that defines the $f_{\rm sat}$ weighting is accounted in the *horizon* reservoir at the *horizon* temperature $T_{\rm AH} = H/2\pi$. By Clausius, heat Q deposited into a reservoir at temperature T adds entropy Q/T. When a halo of mass M virialises it releases gravitational binding energy $E_{\rm bind} \propto M^{5/3}$ (NFW; $R_{\rm vir} \propto M^{1/3}$; verified slope 1.64). Under the SEDE prescription this released binding energy is accounted in the horizon entropy ledger at the horizon temperature $T_{\rm AH}$ (we do not claim a demonstrated transport mechanism), so the entropy assigned in this ledger is

$$\Delta S_{AH} = \frac{E_{\rm bind}}{T_{AH}} \propto \frac{M^{5/3}}{H}.$$

Because $T_{\rm AH} \propto H$ is identical for every halo at a given epoch (mass-independent), the per-halo mass weight is $M^{5/3}$: p = 5/3. With $\Sigma_{\rm S}$=∫ $M^{5/3}(dn/d$ ln M)d ln M and a Sheth–Tormen mass function (COLOSSUS, EH98 transfer, M ∈ [10^10, 10^16] $M_\odot$),

$$\gamma = \frac{d\ln\Sigma_S^{(5/3)}}{d\ln\sigma_8} = 1.4964 \approx 1.50.$$

The earlier p = 1 (Result 4B) divided $E_{\rm bind}$ by the *halo's* virial temperature $T_{\rm vir} \propto M^{2/3}$, which is the entropy of the halo's own internal state — correct for the halo, wrong for the entropy added to the cosmic horizon (where $f_{\rm sat}$ lives). The heat is the same; the reservoir, and hence the temperature, is the horizon's. ■

## A.4 Result 5 — the structural-EOS diagnostic (the $w_0$ side-note, not canonical w(z))

Claim (structural EOS). $w_{\rm struct}(0) = -1 + \gamma/[3(e^\gamma - 1)]$, well-approximated by the closed-form algebraic expression $w_{\rm alg} = (4\Omega_{\rm m}/3 - 1)/(1 - \Omega_{\rm m})$. (This is the side-note structural estimate of §3.2, not the model's prediction, which is the canonical crossing w(z).)

Proof. From $\rho_{\rm DE}(z) = \rho_{\rm crit}$,0 $\Omega_{\rm DE0}$ $f_{\rm sat}(z)$ (A.1) with $f_{\rm sat}(x)$, x = D², the structural EOS in the variance variable is $w_{\rm struct}(0) = -1 + (1/3)$ $df_{\rm sat}/dx$|_{x=1}. Differentiating the Theorem-3 form,

$$\left.\frac{df_{\rm sat}}{dx}\right|_{x=1} = \frac{\gamma e^{-\gamma}}{1 - e^{-\gamma}} = \frac{\gamma}{e^\gamma - 1},$$

so $w_{\rm struct}(0) = -1+\gamma/[3(e^{\gamma}-1)]$. At γ = 1.4964 this is 0.432, giving $w_{\rm struct}(0) = -0.856$ (0.33σ from DESI DR2). Numerically $\gamma/(e^{\gamma}-1) = 0.432 \approx \Omega_{\rm m}/(1-\Omega_{\rm m}) = 0.451$ at the observed parameters, so the structural value is approximated to ~4% by the closed-form expression $w_{\rm alg} = (4\Omega_{\rm m}/3-1)/(1-\Omega_{\rm m}) = -0.849$ (0.21σ from DESI). This is a numerical approximation in the x = $D^2$ convention, not an exact identity. (The interacting-fluid effective EOS, from Raychaudhuri + Friedmann at z = 0, is phantom, $w_{\rm fluid}(0) \approx -1.15$, because $f_{\rm sat}$ is still rising today; the canonical λ = 1/2 background realises this as a crossing of −1, §5.1.) ■

## A.5 Result 8 — the H-coupling λ = 1 − Δ/2

Claim. With the Barrow entropy $S = (A/A_0)^{1+\Delta/2}$, the conjugate horizon-fluid ansatz gives $\rho_{\rm DE} \propto H^{2\lambda} f_{\rm sat}$ with λ = 1 − Δ/2.

Proof. A quantum-gravitationally rough horizon carries the Barrow entropy $S = (A/A_0)^{1+\Delta/2}$, Δ ∈ [0,1], in place of the area law. With A = 4π/$H^2$ and $V_{\rm AH} \propto H^{-3}$, the Barrow-deformed entropy density (written $s_\Delta$ to distinguish it from the Bekenstein–Hawking area-law density $s_{\rm AH} \propto H$) is

$$s_\Delta = \frac{S}{V_{AH}} \propto \frac{(H^{-2})^{1+\Delta/2}}{H^{-3}} = H^{1-\Delta},$$

$$\rho_{\rm DE} = T_{AH}\, s_\Delta\, f_{\rm sat} \propto H \cdot H^{1-\Delta} \cdot f_{\rm sat} = H^{2-\Delta} f_{\rm sat} \equiv H^{2\lambda} f_{\rm sat}, \qquad \boxed{\lambda = 1 - \tfrac{\Delta}{2}.}$$

This is Barrow holographic dark energy with the Hubble IR cutoff ($\rho_{\rm DE} \propto L^{\Delta-2}$, L = 1/H). The endpoints are Δ = 0 ⟹ λ = 1 (the area-law branch, disfavoured) and Δ = 1 ⟹ λ = 1/2 (the volume-law postulate, A.6). λ is therefore not a free parameter; it is set by the Barrow deformation. Scope: the Barrow exponent multiplies only the $f_{\rm sat}$-gated $\rho_{\rm DE}$, *not* the gravitational sector, so H(z) at BBN equals the standard matter+radiation value to ~8 figures ($\Omega_{\rm DE}/\rho_{\rm tot}$ ~ $10^{-16}$ at z ~ 4×$10^9$); the modified-gravity BBN bound Δ $\lesssim 10^{-4}$ does not apply, and constant Δ = 1 is BBN-safe. ■

## A.6 Motivation 9 — Δ = 1 from a geometric ceiling and the GSL

Claim. Δ = 1 (the volume-law endpoint) is *strongly motivated* — not fitted to cosmological data — by two independent principles; it is the central postulate (§8.4), not a microscopic derivation.

Pillar 1 — geometric ceiling. The Barrow deformation is the fractal (Hausdorff) dimension of the horizon surface, $d_{\rm H} = 2 + \Delta$. A 2-surface embedded in 3-space cannot exceed the space-filling dimension $d_{\rm H} = 3$, so $\Delta \le 1$, with Δ = 1 the extremum where the area-measure becomes a volume-measure ($S \propto A^{3/2} \propto R^3$). This is Barrow's defining range.

Pillar 2 — thermodynamic attractor. Three premises: (P1) the DE sector is purely entropic — SEDE posits $\rho_{\rm DE} = T_{\rm AH}\, s_{\rm grav}$, so the free-energy density $F = \rho_{\rm DE} - T_{\rm AH}\, s_{\rm grav} = 0$ identically; the governing potential is therefore the *entropy*, not F = E − TS (this is the load-bearing step: with a nonzero Δ-dependent free energy, equilibrium would extremise F and Δ = 1 would not follow). (P2) the generalised second law: $S_{\rm total} = S_\Delta + S_{\rm matter}$ is non-decreasing, $S_{\rm matter}$ independent of Δ. (P3) Δ is a thermalising horizon degree of freedom (the standard Jacobson–Padmanabhan assumption). By (P1)–(P2), maximising $S_{\rm total}$ over Δ reduces to maximising $S_\Delta = (A/A_0)^{1+\Delta/2}$, whose derivative is

$$\frac{dS_\Delta}{d\Delta} = \frac{S_\Delta}{2}\ln\frac{A}{A_0} + [\text{back-reaction through } A(\Delta)].$$

The explicit term carries ln($A/A_0$) ≈ ln($10^{122}$) ≈ 280 > 0 (a vastly super-Planckian horizon). The back-reaction (Δ shifts $\rho_{\rm DE} \propto H^{2-\Delta}$, hence H, hence A) cannot flip the sign: it *vanishes* at z = 0 (H = $H_0$ ⟹ ln H = 0 ⟹ ∂_Δ $H^{2-\Delta} = 0$) and is ~1/280-suppressed elsewhere. Hence $dS_\Delta/d\Delta > 0$ on all of [0,1]; a strictly increasing function on a closed interval attains its maximum at the right endpoint. With the ceiling $\Delta \le 1$ (Pillar 1) the constrained maximum is uniquely Δ = 1, and by (P3) a thermalising horizon relaxes to it.

Conclusion. Under the additional assumptions that Δ is a thermalising horizon degree of freedom (P3) and that entropy maximisation over the Barrow family is the correct variational principle (P1), the GSL argument is a consistency check — the endpoint Δ = 1 is a GSL *fixed point* — rather than a selection principle: the geometric ceiling $d_{\mathrm{H}} = 3$ supplies the bound $\Delta \leq 1$, and saturation of that cap *is* the volume-law postulate in effective form (via $\lambda = 1 - \Delta/2$ this gives $\lambda = 1/2$). These assumptions motivate but do not microscopically derive the endpoint (§8.4). Because $S_\Delta$ is monotone in Δ, the same argument favours Δ = 1 at all epochs (the adopted constant Δ). The current conditional diagnostics prefer values near $\Delta(0) \approx 1.0$–1.1 (§5.6), consistent with the endpoint. This is a strong motivation, not a derivation: it reduces Δ = 1 from a fitted coupling to a GSL-equilibrium extremum under the stated premises, leaving the microscopic origin open. ■

### A.7 The remaining repository items (catalogue)

Result 2 (reheating master equation): the same logistic $df_{\mathrm{sat}}/dt = H\, f_{\mathrm{sat}}(1 - f_{\mathrm{sat}}) - \Gamma\, f_{\mathrm{sat}}$ governs the *early* $f_{\mathrm{sat}}$ (inflaton→radiation), giving the U-shape of Result 12. Result 4 / 4B [superseded by 4C]: the $M^{5/3}$ energy weight, and the p = 1 virial-temperature mis-step. Result 5D: the (1 − ε/2) "λ = 1 cousin" background ($w_0 = -0.85$, no crossing) vs the canonical λ = 1/2 crossing background. Result 6 / 6B: the smooth-DE growth validation and the $c_{\mathrm{s}}^2 = 1$ closure ($f_{\mathrm{sat}}$ a background functional). Result 10: the QG chain and the volume-law postulate. Result 11: the seam = BBN bound = modgrav/holographic fork. Result 12: the $f_{\mathrm{sat}}$ U-shape (inflation and dark energy as the two de Sitter brackets). Result 13: $1 + w = (1/3)[2\lambda\varepsilon - d \ln f_{\mathrm{sat}}/d \ln a]$, the EOS as horizon-entropy production. Full statements in `sede/theory.py`; each is exercised by a numbered check in App. B.

## Appendix B — The verification suite

The repository ships a `sede/verification.py` suite of 63 numerical checks (V1–V63) covering: the fixed-point solution (closed-loop H(z) to $10^{-13}$ over $z \in [0, 10^6]$), BBN-safety ($\Delta N_{\mathrm{eff}} = \Delta Y_{\mathrm{p}} = 0$), the CAMB≡CLASS $r_{\mathrm{drag}}$ agreement ($5\times10^{-5}$), the CLASS perturbation validation (0.1%), the GSL along cosmic history, the fixed γ prescription and the smooth-DE $c_{\mathrm{s}}^2 = 1$ closure, the horizon-dimension scaling $\lambda(d_{\mathrm{H}}) = 2 - d_{\mathrm{H}}/2$ (V59), the §8.5 state/counting diagnostics (SYK scrambling V60; causal-set dof counting in Minkowski + de Sitter V61; the entropy-bound audit V62; the tensor-network realisability test V63), and the cross-horizon and quantum-sector consistency checks. `reproduce_all.py` regenerates every number and figure in this paper from the staged drivers.

## Appendix C — Data vectors, covariances, and the fσ8 audit

The full data vectors and covariance matrices, the positive-definiteness checks, the benign Pantheon+ duplicate-CID treatment, and the fσ8 data audit (the small set of mis-entered growth points corrected to their published Gold-2018 values, and the SH0ES value correction) — all reproduced from `data/` and documented in `DATA_AUDIT.md`.

## Appendix D — False-preference calibration methodology

The parametric-bootstrap procedure of §5.2: generation of ΛCDM-truth mocks with all probes (including the compressed CMB $R/l_{\mathrm{A}}$ and SH0ES) re-drawn self-consistently; the refit of both models to each mock; the null distribution of $\Delta\chi^2$(SEDE − ΛCDM) (mean +0.37, 2000 mocks); the $p \approx 0.004$ (95% UL 0.006); the documented bug (CMB held fixed → biased null −2.87, $p \approx 0.5$) and its fix; and the sibling-pipeline cross-check (~2σ, CMB-distance-engine-dependent). The pre-registration sha256 and contents are listed. The standalone `PREREGISTRATION.md` additionally records the triggered CMB follow-ups and their status (§E: full non-lite plik run, $\Delta\chi^2$=−0.33; ACT DR6 primary formally deferred) and the locked E1 structure-sourcing decision rule (§F: data vector + CONFIRM/REFUTE thresholds). The no-SH0ES robustness (§5.4 ii′; $\Delta\chi^2$=−1.75, ln B=+0.95, p=0.030) is reproduced by `run_no_shoes_robustness.py`.

# Appendix E — The QG derivation ledger

The full chain from §8 with each link labelled by status: - Jacobson–Cai–Kim (MOTIVATED / ANSATZ): Clausius horizon thermodynamics motivates the horizon-fluid construction and supplies the conservation/dynamical logic (δQ = T dS $\Longrightarrow$ Friedmann); the *absolute* identification $\rho_{\rm DE} = T \cdot s$ is the SEDE ansatz (A.1), not a theorem. - Non-extensivity $\Longrightarrow$ Tsallis–Cirto/Barrow power law (MOTIVATED): the weakest link. - Second-law / GSL $\Longrightarrow$ volume-law endpoint Δ = 1 (MOTIVATED, given the volume-law thermalisation premise): the geometric ceiling caps $\Delta \leq 1$ and the GSL is *consistent with* (a fixed point at) the saturated endpoint — a consistency check, not a selection principle; a microscopic derivation remains OPEN. - CKN scale (RIGOROUS BOUND, normalised by flatness): the holographic energy bound gives the horizon-scale upper bound; the amplitude is normalised by flatness (§8.2). - The one irreducible postulate (OPEN): a microscopic state count delivering the volume law.

The perturbative-QG (log-correction) vs thermalised (volume-law) regimes are reconciled as different *states*, not a contradiction. This ledger matches §8.4 and Appendix F.

# Appendix F — The energy and entropy budget

This appendix records the back-of-envelope that the energy-conservation framing of §1–§2 turns on, and the resulting ledger of what is derived, normalised, and open.

Energy budget. Dark energy today is $\rho_{\rm DE}$ c$^2$ $\approx$ 0.7 $\rho_{\rm crit}$ c$^2$ $\approx$ 6×10$^{-10}$ J/m$^3$. The gravitational binding energy released by structure is bounded by $|E_{\rm bind}|$/Mc$^2$ ~ σ$^2$/c$^2$ per virialised system, with σ ~ 10$^3$ km/s for clusters (σ$^2$/c$^2$ ~ 10$^{-5}$) and ~200 km/s for galaxies (~10$^{-7}$); even taking all of $\Omega_{\rm m}$ collapsed, $\rho_{\rm bind}$ c$^2$ $\lesssim$ 10$^{-5}$ · 0.3 · 8.5×10$^{-10}$ ~ 10$^{-15}$ J/m$^3$. Hence $\rho_{\rm bind}/\rho_{\rm DE}$ ~ 10$^{-6}$: binding energy is six orders of magnitude too small to *be* the dark energy. A literal matter$\to$dark-energy transfer is excluded independently — moving $O(\rho_{\rm crit})$ out of matter over a Hubble time would spoil $\rho_{\rm m} \propto a^{-3}$ and with it BBN and the CMB.

Entropy budget. Deposited at the cold horizon temperature, the binding entropy is $\Delta S_{\rm AH}$= $E_{\rm bind}/T_{\rm AH}$; with $T_{\rm AH} = \hbar H/2\pi$ this comes to $\Delta S_{\rm AH}$ ~ 10$^{-7}$ $S_{\rm AH}$, i.e. ~10$^{-7}$ of the horizon entropy. So binding-entropy deposition does not, by itself, fill $f_{\rm sat}$ to O(1); it sets the *relative* z-dependence (the weight p = 5/3 $\to$ γ), while the normalisation $f_{\rm sat}(0) = 1$ is set by the horizon capacity. This is the same family as the $(M_{\rm P}/H)^{\Delta}$ ~ 10$^{61}$ "seam" (Result 11), resolved by the holographic-DE scope (Barrow acts only on the $f_{\rm sat}$-gated dark-energy fluid).

The ledger.

| quantity | status | set by |
|---|---|---|
| magnitude $\rho_{\rm DE}$ ~ $\rho_{\rm crit}$ | bounded + normalised | CKN holographic upper bound (§8.2), amplitude fixed by flatness (O(1) saturation of the bound assumed) |
| H-scaling λ = 1/2 | postulate | volume-law horizon entropy $S_{\rm grav} \propto V_{\rm AH}$ (§8.1) |
| shape γ ≈ 1.5 | fixed prescription | halo-binding-energy weighting $E_{\rm bind} \propto M^{5/3} \to$ p = 5/3 (§3.1) |
| sound speed $c_{\rm s}^2 = 1$ | closure | minimal smooth-DE treatment, $f_{\rm sat}$ a background functional (§3.4) |
| normalisation $f_{\rm sat}(0) = 1$ | imposed | present-day horizon-entropy normalisation |
| volume-law horizon (Δ = 1 microscopically) | open | the one irreducible postulate (§8.4) |

The honest reading: none of these is a parameter *fitted to cosmological data*, but neither are they all "derived from first principles" — the magnitude is a holographic bound normalised by flatness (as ΛCDM normalises $\Omega_{\Lambda}0$), the H-scaling is the central volume-law postulate, γ is a fixed halo prescription, and $c_{\rm s}^2 = 1$ is a modelling closure. Binding energy enters SEDE entropically, fixing the *shape* of the gate; the dark energy's *magnitude* is the horizon's own thermal scale, not energy borrowed from structure. The single irreducible foundational input is the microscopic volume-law postulate.

## Appendix G — Supporting motivations and numerical probes for the volume-law postulate

This appendix collects the material demoted from §8 to keep the main line focused: three further consistency *motivations* for the volume endpoint (§8.1) and the *numerical probes* of the state and counting halves of the postulate (§8.5). None is a derivation; each is a consistency check that makes the postulate less *ad hoc* without selecting it. (The CHR near-critical layer, the cross-horizon black-hole reading, and the Verlinde dark-matter connection are developed in a separate companion paper.)

### G.1 Three further consistency motivations for the volume endpoint

*(supplementing the geometric ceiling, the CKN scale/form split, and the direct area-law disfavouring of §8.1)*

- A driven steady state. The horizon need not self-thermalise — its scrambling time ($\sim$ ln S / H $\approx$ 280/H) far exceeds its age. Instead the structure-growth gate (the rise of $f_{\rm sat}$) *drives* it off the ground state; with a scrambling-limited relaxation rate the volume-law state is *maintained* as a non-equilibrium steady state throughout structure formation (the long scrambling time becomes an asset). The volume law is then a fixed point of SEDE's own dynamics, not an external input.
- A roughening universality class (developed with the sibling $SEDE_{\rm V2}$ collaboration). Modelling the horizon as a stochastic growing surface, $\partial h/\partial t = \nu\nabla^2 h + (\lambda_{\rm K}/2)(\nabla h)^2 + \eta$, the deformation is fixed by universality class: the thermal/linear Edwards–Wilkinson class ($\lambda_{\rm K} = 0$) flows to the volume endpoint and corresponds to SEDE's purely-entropic F = 0 condition, whereas a generic nonlinear (KPZ) surface gives a sub-volume value. This is the most speculative of the motivations and is included only as such.
- An independent emergent-gravity proposal (Verlinde 2016). In Verlinde's programme the de Sitter vacuum carries a volume-law contribution to its entanglement entropy, $S \propto V_{\rm H}$, in addition to the horizon area law — the same super-area scaling SEDE adopts at Δ = 1. A second, independently-motivated route arriving at a volume-law entropy makes the postulate less *ad hoc*; it remains a motivation, not a proof, since Verlinde's volume-law derivation is itself heuristic.

### G.2 The state is maximally entangled (Majorana-SYK diagonalisation)

The *state* half of the postulate (§8.5) — that the horizon dof are maximally entangled (thermal), not in a low-entanglement ground state — is settled and not peculiar to SEDE. Exact diagonalisation of the Majorana Sachdev–Ye–Kitaev model [Sachdev & Ye 1993; Kitaev 2015], the canonical chaotic, black-hole-dual system saturating the Maldacena–Shenker–Stanford bound $\lambda_{\rm L} \leq 2\pi T$ [arXiv:1503.01409] (`run_syk_scrambling.py`, V60), confirms: the gap ratio reproduces Wigner–Dyson statistics with the correct N-mod-8 symmetry class [García-García & Verbaarschot 2016; Cotler et al 2017; Atas et al 2013], eigenstates saturate the Page value [Page 1993; Vidmar & Rigol 2017], and the OTOC scrambles completely — while an integrable control does none of these. But this does not fix Δ: a black hole is a *maximal* scrambler yet area-law (S = A/4, reproduced by string microstate counts), and SYK, being all-to-all (geometry-free), cannot even pose the counting question.

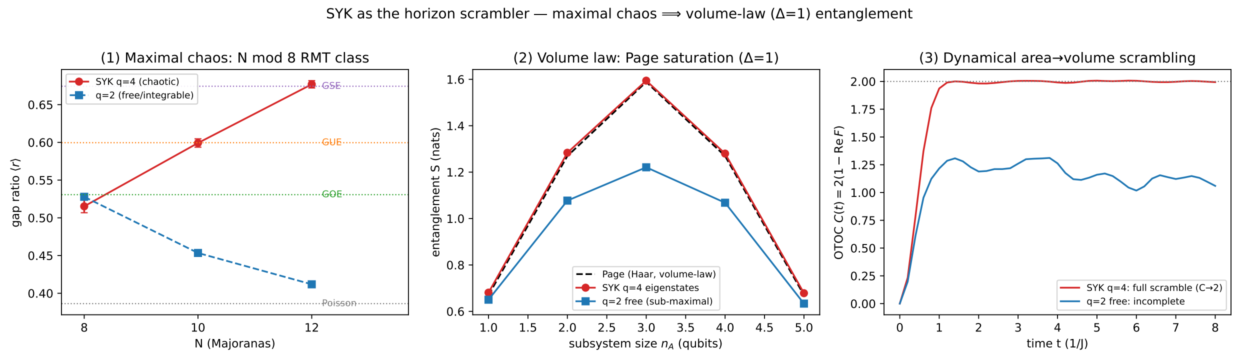


Figure G1. SYK as the horizon scrambler (q=4 chaotic vs q=2 integrable control). *Left:* gap ratio climbs through the N-mod-8 RMT classes (GOE→GUE→GSE) — the certificate of maximal chaos. *Centre:* eigenstate entanglement saturates the Page value. *Right:* the OTOC scrambles completely (C→2). This fixes the *state*, not the counting.

### G.3 The counting question — numerical probes (allowed, realisable, not selected)

The *counting* half (§8.5) — does the horizon dof number grow as area ($N \propto R^{d-2}$, Δ=0) or bulk volume ($N \propto R^{d-1}$, Δ=1)? — is where SEDE's postulate lives. Three numerical probes (`run_entropy_bounds.py`, `run_causal_set.py`, `run_tensor_network.py`; V62, V61, V63):

| question | probe | result |
|---|---|---|
| Allowed? by the established entropy bounds | volume entropy vs the smooth-area bounds | S ∝ $R^3$ overshoots the holographic/Bousso/Bekenstein bound A/4 by ∝ R (~$10^{61}$ at the cosmic horizon); consistent only as a genuine $d_{\mathrm{H}} = 3$ fractal horizon (true area ∝ $R^3 \Longrightarrow S = A_{\mathrm{fractal}}/4$ *exactly* saturates the bound) |
| Selected? by quantum gravity | causal-set sprinkle, Minkowski and de Sitter | both scalings present; the canonical horizon object — causal *links* — is area ($R^{d-2}$, the Bekenstein–Hawking recovery), in flat space and, because dS is conformally flat, identically in de Sitter. Volume is not the default. (measured: bulk count 1.92≈2 $\Longrightarrow$ Δ=1; link count 0.98≈1 $\Longrightarrow$ Δ=0, in 2+1D) |
| Realisable? in a holographic code | Ryu–Takayanagi min-cut, local vs nonlocal | local (lattice) connectivity $\Longrightarrow$ area law (exponent 1.0); nonlocal (expander) connectivity $\Longrightarrow$ volume law (exponent 1.9). Δ=1 is realisable; its required ingredient is *nonlocal* horizon connectivity (the all-to-all structure of SYK). |

The pattern: volume-counting is allowed (not by evading the bounds but by the horizon being genuinely space-filling — the postulate itself), realisable (a nonlocal holographic code produces $S \propto R^{d-1}$ exactly), but not selected (the frameworks we can compute in default to area for the canonical horizon object, and a black hole is area-law). No consistency argument available to us *derives* the volume count; equally, none *excludes* it. The two deciders are stated in §8.5: the dS static-patch Hilbert-space dimension (theoretical) and the deformation $\Delta$ (empirical, §5.6/§6).